KU LEUVEN

FACULTY OF SCIENCE

Maximising Edge Density in Graphs Subject to Connectivity Constraints

Erdős Problem 915, from Proof to Search

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Thesis presented in
fulfilment of the requirements
for the degree of Master of Science
in Mathematics

Academic year 2026–2027

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Foreword

This thesis asks how many edges a network can have when no two vertices may be joined by too many independent routes. The starting point is Erdős Problem 915, which asks this for edge-disjoint paths in simple graphs. Chapter 1 states the problem and its variants, and collects what is proved. Chapter 2 explains how a program can measure connectivity and search for dense examples. Chapter 3 summarises the results and lists the open problems. The appendix contains the proofs.

Contribution Statement

I carried out this project with Stijn Cambie. I used artificial intelligence to write code, to suggest proofs, to help check them, to format LaTeX and to check spelling. Results from the literature are cited where they are used. The code is available in the repository [1].

Every theorem and lemma is cited from the literature where needed.

Area	Main results
Undirected graphs	A proof of the edge bounds with the Gomory–Hu tree, and extremal graphs for every $n \geq m$.
Hypergraphs, edge version	An upper bound for every r-uniform hypergraph, attained by simple hypergraphs on a stated range.
Hypergraphs, vertex version	The exact value at $m = 2$, and an upper bound at $m = 3$ that is attained on stated ranges.
Directed multigraphs	The exact value for every n and m, and the extremal multigraphs for $n \geq \max(8, 2m)$.
Simple digraphs	The exact value at $m = 2$, for both separations.
Multigraphs, vertex version	The exact value at $m = 2$ and $m = 3$.
Directed graphs and hypergraphs	Upper bounds with the right leading term, $n^2/4$ for simple digraphs and $(m-1)n^2/(4(r-1))$ for directed hypergraphs, valid for every n.

Short Summary

Erdős Problem 915 asks how many edges a simple graph on n vertices can have if no pair is joined by m edge-disjoint paths [2, 3]. Writing the maximum as $\ell_m(n)$, Mader's bound is

$$\ell_m(n) \le \left\lfloor \frac{m(n-1)}{2} \right\rfloor .$$

We prove this bound with the Gomory–Hu tree and give graphs that reach it for every $n \ge m \ge 2$.

The thesis also studies the problem in eight models. Edges may be directed or not, may be repeated or not, and may join two vertices or more (hypergraphs). In each model we can ask for edge-disjoint routes or for vertex-disjoint routes, which gives sixteen problems.

The proved results include:

- Multigraphs: at most $(m-1)(n-1)$ edges.
- r-uniform hypergraphs: at most $\lfloor (m-1)(n-1)/(r-1) \rfloor$ hyperedges.
- Hypergraphs with vertex-disjoint routes: exactly $\lfloor (n-1)/(r-1) \rfloor$ at $m = 2$, and at most $\lfloor 2(n-1)/(r-1) \rfloor$ at $m = 3$.
- Directed multigraphs: exactly $(m-1)\max\bigl(2(n-1), \lfloor n^2/4 \rfloor\bigr)$ arcs.
- Simple digraphs: $n^2/4 + \Theta_m(n)$ arcs for fixed $m \ge 3$.

The exact value for simple digraphs is open for $m \ge 3$.

Chapter 2 explains how a program measures connectivity with maximum flow and searches for dense examples. No result depends on this program.

Abbreviations and Symbols

Symbol	Meaning
$G = (V, E)$, $D = (V, A)$	vertex set V, with edge multiset E or arc multiset A
n, m, r	vertex count, forbidden-connectivity threshold, hyperedge size
$\lambda_G(u, v)$	maximum number of edge-disjoint u–v paths
$\kappa_G(u, v)$	maximum number of incidence-separated u–v paths
$\lambda_G^{\max}$, $\kappa_G^{\max}$	maximum local connectivity over distinct vertex pairs, ordered in directed models
$\ell_m(n)$, $L_m(n)$	simple and multigraph undirected edge maxima
$k_m(n)$, $K_m(n)$	simple and multigraph undirected vertex maxima
$\ell_m^{\mathrm{dir}}(n)$, $L_m^{\mathrm{dir}}(n)$	simple and multigraph directed arc maxima
$k_m^{\mathrm{dir}}(n)$, $K_m^{\mathrm{dir}}(n)$	simple and multigraph directed vertex maxima
$\ell_m^{(r)}(n)$, $L_m^{(r)}(n)$	simple and multi r-uniform hypergraph edge maxima
$k_m^{(r)}(n)$, $K_m^{(r)}(n)$	simple and multi r-uniform hypergraph vertex maxima
$M(n)$	$\max(2(n-1), \lfloor n^2/4 \rfloor)$, the directed arc maximum at $m = 2$
$B_{p,q}$	complete one-way bipartite digraph, from the p-part to the q-part
$\mu(u, v)$	edge or arc multiplicity
$h_m(b)$	maximum edge count of a feasible simple block on b vertices
$W_m(G_0)$	maximum feasible total multiplicity over the simple support G_0
$g_m(b)$	maximum W_m of a feasible block on b vertices
c_m	limiting rate $\lim_{n\to\infty} k_m(n)/(n-1)$ (Theorem A.11)

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Chapter 1

The Problem and Its Variants

This thesis asks how many connections a network can have before some pair of points has too many independent routes between them. This chapter defines the problem and its variants, and states the results that are proved.

1.1 Graphs and routes

A *multigraph* G consists of a finite set $V(G)$ of *vertices* and a multiset $E(G)$ of *edges*. Each edge joins two different vertices, and edges can have identical copies. The number of such identical edges between u and v is the multiplicity $\mu(u, v)$. If $\mu(u, v) \leq 1$ for every pair, then G is a *simple graph*. The *size* of G is its number of edges, counted with multiplicity: $|E(G)| = \sum_{\{u,v\}} \mu(u, v)$. This thesis tries to maximize the size. Drawing vertices as dots and edges as lines is the standard way to visualize graphs.

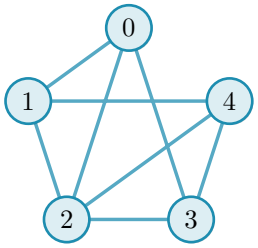


Figure 1.1. The simple graph $K_5 - \{04, 13\}$ has five vertices and eight edges, where K_n is the complete graph on n vertices.

We write $n := |V(G)|$. The *degree* of a vertex is the number of edges at it. From now on, graphs are simple unless we say otherwise. A *path* from u to v is a sequence $v_0, e_1, v_1, \ldots, e_q, v_q$ of distinct vertices v_i and edges e_i, where $v_0 = u$, $v_q = v$ and e_i joins v_{i-1} and v_i. In a multigraph, two parallel edges give two different paths of length one. In a simple graph we can write a path as a list of vertices. A graph is *connected* if every two vertices are joined by a path. A *tree* is a connected graph in which every two vertices are joined by exactly one path. A tree on n vertices has $n - 1$ edges [4]. A *spanning tree* of a graph G is a tree on all vertices of G that uses only edges of G. Two u–v paths are *edge-disjoint* if they have no edge in common. The *local edge-connectivity* $\lambda_G(u, v)$ is the largest number of pairwise edge-disjoint u–v paths. Its largest value over all pairs is

$$\lambda_G^{\max} = \max_{u \neq v} \lambda_G(u, v).$$

If G has fewer than two vertices, we set $\lambda_G^{\max} = 0$.

Theorem 1.1 (Menger [5, 6]). *Let u and v be distinct vertices of a multigraph G. The largest number of pairwise edge-disjoint u–v paths equals the smallest number of edges whose removal leaves no u–v path.*

Figure 1.2 shows an example.

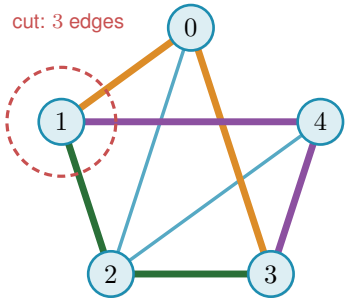


Figure 1.2. Menger's theorem on Figure 1.1: three edge-disjoint 1–3 routes (coloured) meet the minimum three-edge cut around vertex 1, so $\lambda(1,3) = 3$.

1.2 Erdős Problem 915

Fix an integer $m \geq 2$. The condition $\lambda_G^{\max} \leq m-1$ means that no two vertices are joined by m edge-disjoint paths. By Menger's theorem, it also means that any two vertices can be separated by removing at most $m-1$ edges. We ask how many edges a graph can have under this condition.

Question 1.2 (Erdős Problem 915 [3]). How many edges can a graph on n vertices have if no two of its vertices are joined by m disjoint paths?

The database [3] notes that the original statement leaves open whether "disjoint" means edge-disjoint or internally vertex-disjoint. This thesis takes edge-disjoint paths as the main question and treats vertex-disjoint paths as a variant (Section 1.3).

We write

$$\ell_m(n) = \max\{\, |E(G)| : |V(G)| = n,\ \lambda_G^{\max} \leq m-1 \,\}$$

for the answer. Mader proved the following upper bound where $\lfloor x \rfloor$ is x rounded down and $\lceil x \rceil$ is x rounded up.

Theorem 1.3 (Mader [7]). *For all $n \geq 2$ and $m \geq 2$,*

$$\ell_m(n) \leq \left\lfloor \frac{m(n-1)}{2} \right\rfloor .$$

Bollobás and Erdős [2] found graphs that attain this bound when $m-1$ divides $n-1$: copies of K_m that share one vertex. For $n \geq m$ the bound is always attained (Theorem A.8). For $2 \leq n < m$ the complete graph K_n satisfies the condition, so $\ell_m(n) = \binom{n}{2}$. This is smaller than $\lfloor m(n-1)/2 \rfloor$, except when $(n,m) = (2,3)$.

Section A.2 proves Theorem 1.3 with the Gomory–Hu tree as alternative independently found. This is a tree on the vertices of G with a weight on each edge. For every pair u, v, the value

$\lambda_G(u, v)$ is the smallest weight on the path from u to v in the tree. So the $\binom{n}{2}$ values $\lambda_G(u, v)$ are stored in only $n - 1$ weights. The edges of the tree need not be edges of G. The proof counts the edges of G in two ways with this tree. The same method proves the bound for multigraphs (Theorem 1.4).

1.3 Variants of the problem

We change four assumptions. The first three give eight graph models. In each model we can ask for edge-disjoint or for vertex-disjoint routes (Section 1.3.1).

The first assumption is the size of an edge. A *hyperedge* is a set of $r \geq 2$ vertices. In an *r-uniform hypergraph* every hyperedge has exactly r vertices. A *Berge path* is a sequence of distinct vertices and distinct hyperedges, alternating, in which each hyperedge contains the vertex before it and the vertex after it. This aligns best with the path definition for multigraphs. The size of a hypergraph is its number of hyperedges, counted with multiplicity.

The second assumption is repetition. A *simple* graph or hypergraph has each edge at most once. A *multigraph* or *multihypergraph* may repeat edges.

The third assumption is direction. A directed edge, or *arc*, is an ordered pair of vertices. A directed hyperedge has one *tail* and $r - 1$ *heads*. A directed Berge route enters a hyperedge at its tail and leaves it at a head. In the figures, one colour is one hyperedge, not a path of ordinary edges. Figure 1.3 shows the eight models.

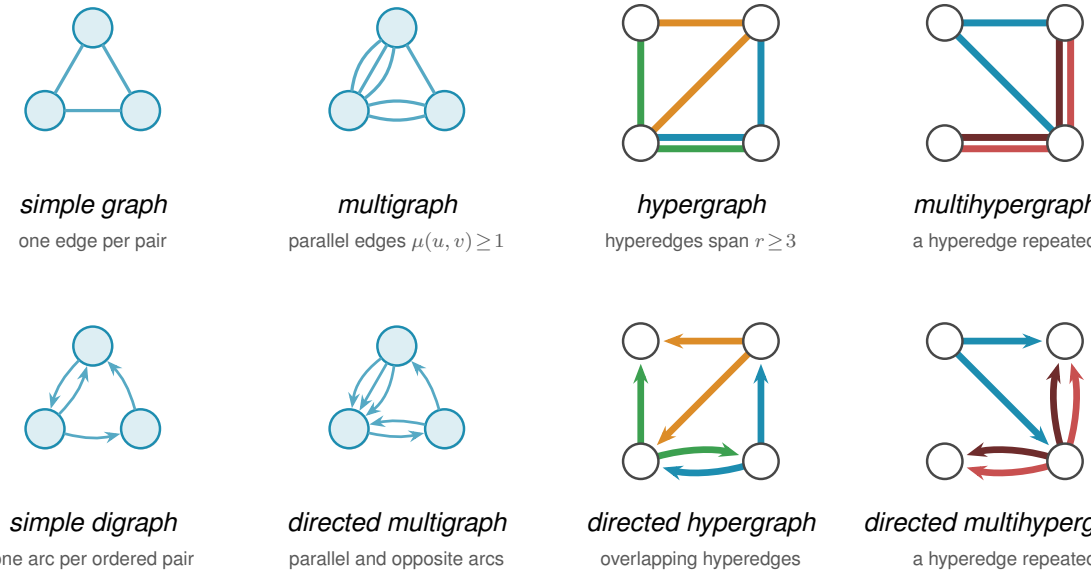


Figure 1.3. The eight models side by side, with the four undirected on top and the four directed below. The simple graph is a triangle, the multigraph has parallel edges, the hypergraph has three hyperedges, and the multihypergraph has one repeated hyperedge. The directed versions have arcs instead of edges, with the same pattern of repetition and hyperedges. In the two multihypergraph panels the two shades of red are the two copies of one hyperedge.

Digraphs have no Gomory–Hu tree, but Menger's theorem holds for them. In a digraph, the *out-degree* $d^+(v)$ is the number of arcs that leave v. The vertices these arcs point to are the *out-neighbours* of v. The *in-degree* $d^-(v)$ is the number of arcs that enter v, and they come from the

in-neighbours of v. The *total degree* is $d(v) = d^+(v) + d^-(v)$. Figure 1.4 shows three arc-disjoint routes from u to v. Removing three arcs cuts all of them, as Menger's theorem predicts. The out-degree of u and the in-degree of v also bound the number of routes.

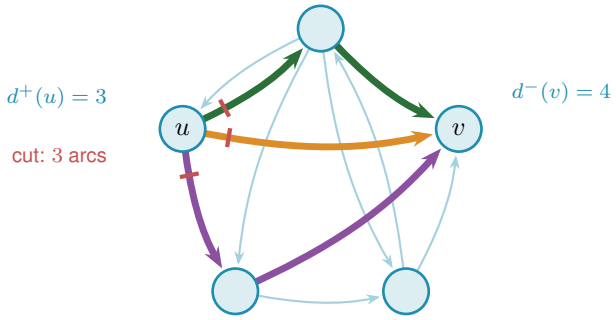


Figure 1.4. Three arc-disjoint $u \to v$ routes (coloured) and the corresponding three-arc directed cut (red) in a *simple* digraph. Thus $\lambda(u, v) = 3$, meeting the degree bound $\min(d^+(u), d^-(v)) = 3$.

The fourth assumption is *separation*. Two u–v paths are *internally vertex-disjoint* if they have no vertex in common except u and v. The largest number of such paths is the *local vertex-connectivity* $\kappa_G(u, v)$. Its largest value over all pairs is $\kappa_G^{\max}$. Internally vertex-disjoint paths are also edge-disjoint, so $\kappa_G(u, v) \leq \lambda_G(u, v)$. Hypergraphs don't always obey this rule. So in order to force the inequality for hypergraphs, internally vertex disjoint paths share neither vertices nor edges. This is just another assumption to simplify the workload and generalize wherever possible. Figure 1.5 shows that the vertex and edge-disjointness can differ.

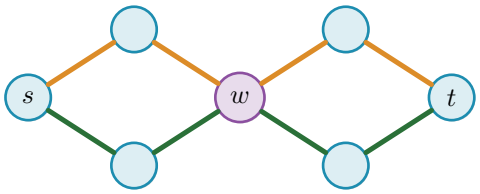


Figure 1.5. The separation axis: two edge-disjoint s–t routes both pass through w, so $\lambda(s, t) = 2$ but $\kappa(s, t) = 1$.

1.3.1 The incidence convention

For multigraphs and hypergraphs we must say when two routes count as vertex-disjoint. This thesis uses the following convention. For simple graphs it gives the usual notion.

The *incidence graph* $I(\mathcal{H})$ has a node for each vertex and a node for each edge copy. An edge copy is joined to the vertices it contains. A route from u to v is a path in $I(\mathcal{H})$. Two routes are *separated* if they have no inner node in common. In the directed case we use the directed incidence graph: the tail has an arc to the edge copy, and the edge copy has an arc to each head. For example, an arc $u \to v$ becomes $u \to e \to v$.

The inner nodes are vertices and edge copies. So separated routes share no inner vertex and no edge copy. This matters for hypergraphs: two Berge routes can use the same hyperedge without meeting in a vertex. Since separated routes share no edge copy, $\kappa \leq \lambda$ also holds here.

In a multigraph, q parallel copies of uv give q separated routes. So the vertex problems

for multigraphs, $K_m(n)$ and $K_m^{\mathrm{dir}}(n)$, differ from the simple ones. In a feasible multigraph no multiplicity is larger than $m-1$, since m parallel copies already give m separated routes.

Table 1.1 lists the notation for all sixteen problems. Figure 3.3 shows what is known about each of them.

Table 1.1. Notation for the extremal functions. The third column uses the incidence convention defined above, in every model.

Model	edge-disjoint routes	vertex-disjoint routes
Simple graph	$\ell_m(n)$	$k_m(n)$
Multigraph	$L_m(n)$	$K_m(n)$
r-uniform simple hypergraph	$\ell_m^{(r)}(n)$	$k_m^{(r)}(n)$
r-uniform multihypergraph	$L_m^{(r)}(n)$	$K_m^{(r)}(n)$
Simple digraph	$\ell_m^{\mathrm{dir}}(n)$	$k_m^{\mathrm{dir}}(n)$
Directed multigraph	$L_m^{\mathrm{dir}}(n)$	$K_m^{\mathrm{dir}}(n)$
Directed r-uniform simple hypergraph	$\ell_m^{\mathrm{dir}\,(r)}(n)$	$k_m^{\mathrm{dir}\,(r)}(n)$
Directed r-uniform multihypergraph	$L_m^{\mathrm{dir}\,(r)}(n)$	$K_m^{\mathrm{dir}\,(r)}(n)$

Menger's theorem, the Gomory–Hu tree and Mader's bound are classical. So are the results of Bártfai, Bollobás, Leonard, Sørensen and Thomassen. The main new results are: the directed values at $m=2$ (Theorems 1.7 and 1.8), the hypergraph edge bound (Theorem 1.9), the exact value for directed multigraphs (Theorem 1.13), the hypergraph vertex problem at $m=2$ and $m=3$ (Theorems 1.10 and 1.11), and the quadratic bounds for directed graphs and hypergraphs (Theorems 1.12 and 1.14).

1.4 Bounds and established results

1.4.1 Undirected graphs

Theorem 1.4. *For $n \geq 1$ and $m \geq 2$, undirected multigraphs satisfy $L_m(n) = (m-1)(n-1)$.*

The proof in Appendix A also uses the Gomory–Hu tree. Equality holds for any tree on the n vertices with every edge repeated $m-1$ times.

In simple graphs, the edge and vertex problems have the same answer for $m \leq 4$.

Theorem 1.5 (Bártfai [8], Bollobás [9], Leonard [10])**.** *For all $n \geq m$ and $2 \leq m \leq 4$,*

$$k_m(n) = \ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor .$$

Since $\kappa \leq \lambda$, every graph with $\lambda^{\max} \leq m-1$ also has $\kappa^{\max} \leq m-1$. So $k_m(n) \geq \ell_m(n)$ (Figure 1.6). Sørensen and Thomassen showed that the two differ for $m=5$.

Theorem 1.6 (Sørensen–Thomassen [11])**.** *For simple graphs,*

$$k_5(n) = \left\lfloor \frac{5(n-1)}{2} \right\rfloor = \ell_5(n) \quad (5 \leq n \leq 13), \qquad k_5(n) = \left\lfloor \frac{8n}{3} \right\rfloor - 4 \quad (n \geq 14).$$

So $k_5(n) > \ell_5(n)$ for every $n \geq 14$. Sørensen and Thomassen count the least number of edges that forces m disjoint paths. This is one more than our maximum (Remark A.10). They give one formula with exceptions at $n = 7$ and $n = 12$. Splitting the statement at $n = 14$, as above, removes these exceptions.

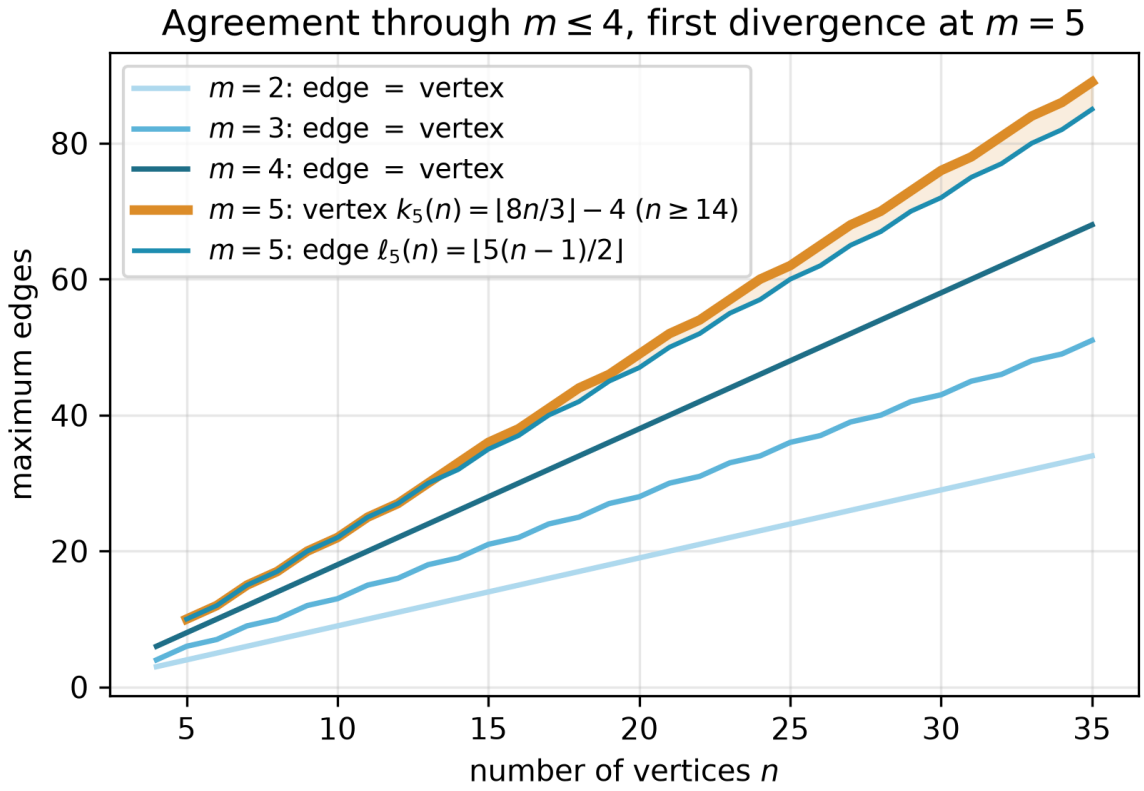


Figure 1.6. The edge and vertex extremal functions for $m = 2, 3, 4$ (blue shades, single curve per m since they agree) and $m = 5$ (thin blue for edge, thick orange for vertex). The vertex function diverges from the edge function at $n = 14$ and grows faster thereafter.

1.4.2 Directed graphs at the first threshold

Theorem 1.7 (Directed arc value at $m = 2$)**.** *For simple digraphs,*

$$\ell_2^{\mathrm{dir}}(n) = \max\left(2(n-1),\ \left\lfloor \frac{n^2}{4} \right\rfloor\right).$$

Two constructions attain the two terms (Figure 1.7). The bidirected star has $2(n-1)$ arcs. The one-way complete bipartite digraph has all arcs from one half of the vertices to the other half, so it has $\lfloor n^2/4 \rfloor$ arcs. In both, every ordered pair is joined by at most one directed path. Section A.3 gives the details.

For the upper bound we use a *reachability skeleton*: a subdigraph in which u can reach v exactly when u can reach v in the whole digraph. Lemma A.17 shows that there is always a skeleton with at most $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ arcs. At $m = 2$, an arc outside the skeleton would give a second route between its ends. So a feasible digraph is its own skeleton.

This proves the bound for vertex separation, and so also for arc separation (Section A.5).

Theorem 1.8 (Directed vertex value at $m = 2$)**.** *For simple digraphs,* $k_2^{\mathrm{dir}}(n) = \ell_2^{\mathrm{dir}}(n)$.

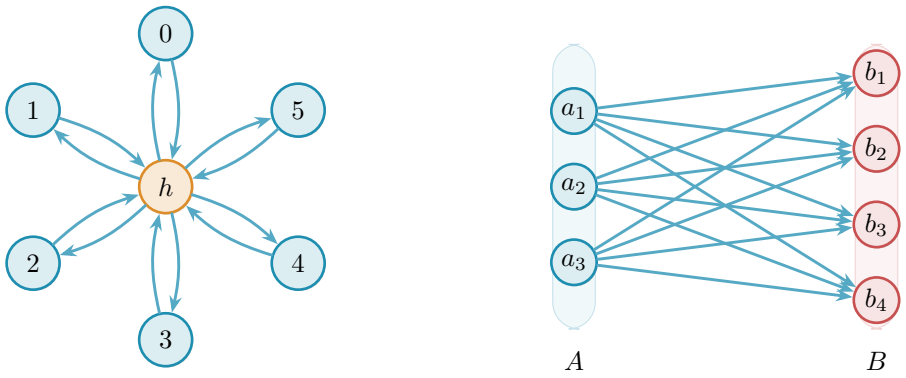


Figure 1.7. The two branches tie at $n = 7$, where each holds 12 arcs at $\lambda^{\max} = 1$. Left: the bidirected star, a hub joined to six spokes both ways, $2(n-1) = 12$. Right: the one-directional complete bipartite digraph $B_{3,4}$, every arc from A to B, $\lfloor n^2/4 \rfloor = 12$.

The star has more arcs for $n < 7$, and the bipartite digraph has more for $n > 7$. So the directed value grows quadratically in n, while the undirected value grows linearly. The reason is that all arcs can point the same way, so no route comes back.

1.4.3 Undirected hypergraphs

Theorem 1.9 (Hypergraph edge bound)**.** *For* $n \geq r \geq 2$ *and* $m \geq 2$*, an* r*-uniform multihypergraph on* n *vertices with* $\lambda^{\max} \leq m-1$ *has size at most* $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$*. There are simple hypergraphs attaining the bound when* $m - 1 \leq \binom{n-2}{r-2}$*, and multihypergraphs attaining it when* $(r-1) \mid n-1$.

Appendix A proves the bound by induction on n: it cuts the hypergraph into two parts. The star hypertree of Figure 1.8 attains the bound at $m = 2$. It has a hub joined to disjoint groups of $r-1$ vertices. For $r \geq 3$, simple hypergraphs attain the bound whenever $m - 1 \leq \binom{n-2}{r-2}$ (Theorem A.35). So for $r \geq 3$, unlike for graphs, repeated edges often do not help. The condition says that there are enough different hyperedges. When it fails, the simple value can be smaller. For example, at $n = r = m = 3$ there is only one possible hyperedge. A simple hypergraph has one hyperedge, while a multihypergraph can have two copies of it (Remark A.36).

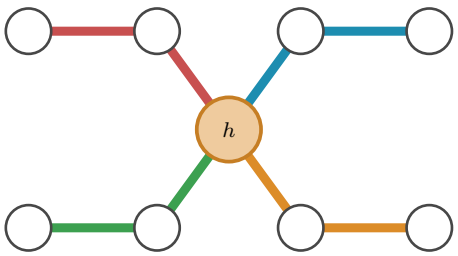


Figure 1.8. The $r = 3$, $n = 9$ star hypertree: four hyperedges meet only at the hub h, attaining $\lfloor (n-1)/(r-1) \rfloor = 4$ with $\lambda^{\max} = 1$.

Theorem 1.10 (Hypergraph vertex value at $m = 2$)**.** *For $r \geq 2$ and $n \geq 1$, the maximum number of hyperedges of an r-uniform multihypergraph on n vertices with $\kappa^{\max} \leq 1$ is*

$$K_2^{(r)}(n) \;=\; \left\lfloor \frac{n-1}{r-1} \right\rfloor,$$

attained by the star hypertree of Figure 1.8, which is simple. In particular $K_2^{(r)}(n) = k_2^{(r)}(n) = \ell_2^{(r)}(n)$: the vertex and edge problems agree at $m = 2$ for every uniformity, and repeated hyperedges never help.

Theorem 1.11 (Hypergraph vertex value at $m = 3$)**.** *For $r \geq 2$ and $n \geq 1$, every r-uniform multihypergraph on n vertices with $\kappa^{\max} \leq 2$ has at most $\left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$ hyperedges. For $2 < r < n$, which is exactly the range where $2 \leq \binom{n-2}{r-2}$, the bound is attained by a simple hypergraph, so*

$$k_3^{(r)}(n) \;=\; \left\lfloor \frac{2(n-1)}{r-1} \right\rfloor \;=\; \ell_3^{(r)}(n).$$

When $(r-1) \mid (n-1)$, a multihypergraph attains it, so then $K_3^{(r)}(n) = \lfloor 2(n-1)/(r-1) \rfloor$ (Proposition A.40).

Both proofs use the incidence graph (Section A.9). At $m = 3$ the proof bounds the number of independent cycles of the incidence graph. We have no such bound for $m \geq 4$.

For $r = 2$ these results are about multigraphs. They give $K_2(n) = n - 1$, attained by trees, and $K_3(n) = 2(n-1)$, attained by a tree with every edge doubled (Corollary A.46). For $m = 5$ the vertex and edge values differ: the thickened theta of Section A.10 gives $K_5(4) \geq 14$, while $L_5(4) = 12$. For $m = 4$ this is open. Section A.10 discusses larger m.

1.4.4 Directed bounds beyond the first threshold

Theorem 1.12 (The quadratic bound with a linear error)**.** *For all $m \geq 2$ and $n \geq 2$, every simple digraph D on n vertices with $\lambda_D^{\max} \leq m-1$, or with $\kappa_D^{\max} \leq m-1$, satisfies*

$$|A(D)| \;\leq\; \left\lfloor \frac{n^2}{4} \right\rfloor \;+\; 4(m-1)(n-1),$$

so for each fixed m,

$$\ell_m^{\mathrm{dir}}(n) \;=\; \frac{n^2}{4} \;+\; \Theta_m(n) \quad \textit{and} \quad k_m^{\mathrm{dir}}(n) \;=\; \frac{n^2}{4} \;+\; \Theta_m(n) \qquad (m \geq 3).$$

Appendix A gives the proof. It counts routes of length two. For an ordered pair (u, v), the routes $u \to x \to v$ with different midpoints x, together with the arc $u \to v$, are disjoint. So a feasible digraph has at most $m-1$ of them. Summing over all pairs bounds $\sum_x d^+(x)\, d^-(x)$. The proof is by induction on n. If some vertex has total degree below $n/2$, we delete it and use the bound on $n-1$ vertices. Otherwise every total degree is at least $n/2$, and the sum above makes $\sum_x \min(d^+(x), d^-(x))$ at most linear in n. At every vertex we then delete its in-arcs or

its out-arcs, whichever are fewer. These deleted arcs give the linear term. What remains has all arcs from one part to the other, so it has at most $\lfloor n^2/4 \rfloor$ arcs. The routes of length two are also internally vertex-disjoint, so the same proof works for vertex separation.

Theorem 1.13 (Directed multigraph value, every n and m)**.** *For all $n \geq 2$ and $m \geq 2$,*

$$L_m^{\mathrm{dir}}(n) \;=\; (m-1)\,M(n), \qquad M(n) := \max\Big(2(n-1),\; \Big\lfloor \frac{n^2}{4} \Big\rfloor\Big).$$

The bound is attained in two ways: by a bidirected tree with every arc taken $m-1$ times, and by the one-way complete bipartite digraph with every arc taken $m-1$ times. The proof works as follows. Every feasible multigraph contains a small reachability skeleton. Removing one copy of each skeleton arc lowers $\lambda(u,v)$ by at least one for every pair with $\lambda(u,v) > 0$. So $m-1$ such removals remove all arcs. For $n \geq \max(8, 2m)$ the bipartite construction is the only extremal multigraph, up to reversing all arcs (Theorem A.32).

A directed hyperedge is *forward* if it has one tail and $r-1$ heads, as above.

Theorem 1.14 (The directed hypergraph constant)**.** *Let $r \geq 2$ and $m \geq 2$. Every forward directed r-uniform multihypergraph $\mathcal{H}$ on $n \geq 2$ vertices with $\kappa_{\mathcal{H}}^{\max} \leq m-1$ satisfies*

$$|E(\mathcal{H})| \;\leq\; \frac{m-1}{r-1}\left\lfloor \frac{n^2}{4} \right\rfloor \;+\; 4(m-1)^2(n-1).$$

Since $\kappa \leq \lambda$, the same bound holds when $\lambda_{\mathcal{H}}^{\max} \leq m-1$. The construction of Proposition A.56 has $(1+o(1))(m-1)n^2/(4(r-1))$ hyperedges, so the leading term is right.

Appendix A gives the proof. It uses the *shadow* digraph, which has an arc $u \to v$ when some hyperedge has tail u and head v. Each hyperedge gives $r-1$ arcs of the shadow, and each arc comes from at most $m-1$ hyperedges. The route condition on the hypergraph limits the routes of length two in the shadow. Lemma A.25 then bounds the number of arcs of the shadow.

At $r = 2$ this also covers directed multigraphs with vertex separation. At $m = 2$ no arc can be repeated, so $K_2^{\mathrm{dir}}(n) = \ell_2^{\mathrm{dir}}(n)$. For fixed m, the extremal multigraphs of Theorem 1.13 are feasible here too, since $\kappa \leq \lambda$. With Theorem 1.14 at $r = 2$ this gives $K_m^{\mathrm{dir}}(n) = (m-1)n^2/4 + O_m(n)$ (Corollary A.53).

Section 3.3.5 compares this with a theorem of Huang and Lyu [12].

Chapter 2

Measuring and Searching by Machine

Chapter 1 stated the problem and the proved results. This chapter explains how a program can measure connectivity in every variant, check small graphs one by one, and search for dense examples. No result in this thesis depends on the output of this program. Figure 2.1 shows the structure of the program.

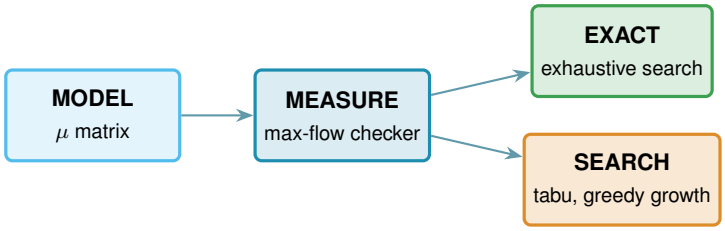


Figure 2.1. One model feeds the max-flow checker. Exhaustive search uses it to find exact values, and heuristic search uses it to find dense examples.

2.1 Modelling

A graph or digraph is stored as a matrix μ of multiplicities. The matrix is symmetric for undirected graphs and has entries 0 and 1 for simple graphs. A hypergraph is stored as a list of hyperedges. In the directed case, each hyperedge also records its tail. Figure 2.2 shows both.

2.2 Measuring

Every search below needs $\lambda^{\max}$ or $\kappa^{\max}$ of a candidate graph. This section explains how the program computes them.

To compute $\lambda(u, v)$ we use a maximum flow. In a network every link has a capacity, and a *maximum flow* is the largest amount that can be sent from u to v. We give each pair of vertices a link with capacity $\mu(u, v)$. In a digraph the link has a direction. In an undirected graph it can be used in both directions. The max-flow min-cut theorem of Ford and Fulkerson [6] says that the maximum flow equals the smallest capacity of a cut. By Menger's theorem, this is $\lambda(u, v)$. So computing a maximum flow computes $\lambda(u, v)$. We call this routine the *checker*.

2.2.1 Transformations

The checker works on directed multigraphs with arc-disjoint routes. The other variants are first turned into such a network. An undirected multigraph becomes a directed one if we replace each edge by two opposite arcs.

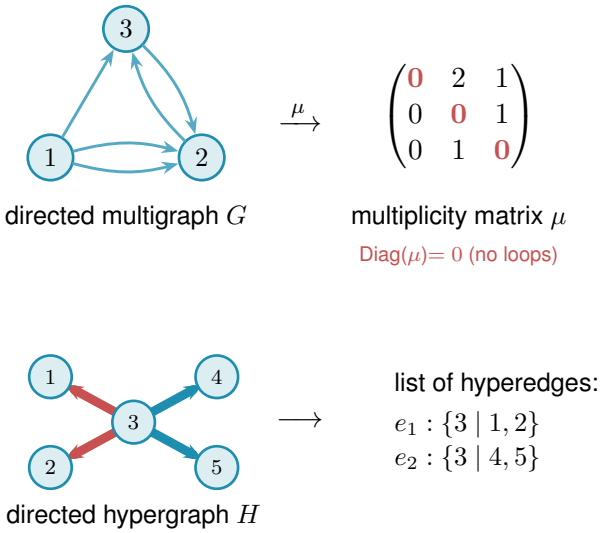


Figure 2.2. The two encodings for directed variants. The multiplicity matrix μ represents a digraph, with μ_{ij} arcs from i to j. A list of hyperedges represents a directed hypergraph and records each tail separately. A multihypergraph repeats entries in that list. A tensor representation would require n^r entries, almost all zero, and is therefore avoided.

A hyperedge becomes a *gate* (Figure 2.4b). The gate has an entry node and an exit node, joined by one arc whose capacity is the multiplicity of the hyperedge. Each vertex of the hyperedge gets an arc of unlimited capacity to the entry node and one from the exit node. So the arc inside the gate is the only bottleneck. For a directed hyperedge, only the tail has an arc to the entry, and only the heads have an arc from the exit (Figure 2.4c). So a route must go from the tail to a head. Theorem A.33 shows that the maximum flow in this network is the number of hyperedge-disjoint Berge routes. Figure 2.3 shows an example.

For vertex separation we split every vertex v into two nodes v^{in} and v^{out}, joined by one arc of capacity one (Figure 2.4a). Each arc $u \to v$ becomes an arc $u^{\text{out}} \to v^{\text{in}}$. A route through v must use the arc from v^{in} to v^{out}, so at most one route can pass through v. So the maximum flow from u^{out} to v^{in} is $\kappa(u, v)$. The arcs between different vertices keep their capacity $\mu(u, v)$, and the gates keep theirs. So parallel copies still give separate routes, as in the incidence convention of Section 1.3.1.

2.3 Checking every graph of a fixed order

To find an exact value, the program must check every graph, or exclude it by a correct pruning rule. A simple graph on n vertices is a choice of $\mu(u, v) \in \{0, 1\}$ for each pair, and a multigraph is a choice in $\{0, \ldots, m-1\}$. A digraph makes this choice for each ordered pair. A hypergraph is a choice of some of the $\binom{n}{r}$ possible hyperedges. A multihypergraph also chooses a multiplicity for each. A directed hypergraph has r possible hyperedges on each r-set, one for each choice of tail, and chooses some of these $r\binom{n}{r}$ hyperedges. The program goes through all choices, checks each one, and keeps the densest feasible graph. If the search is stopped early, it gives only a lower bound.

The number of vertices matters most. The number of candidates is (values per entry) to the

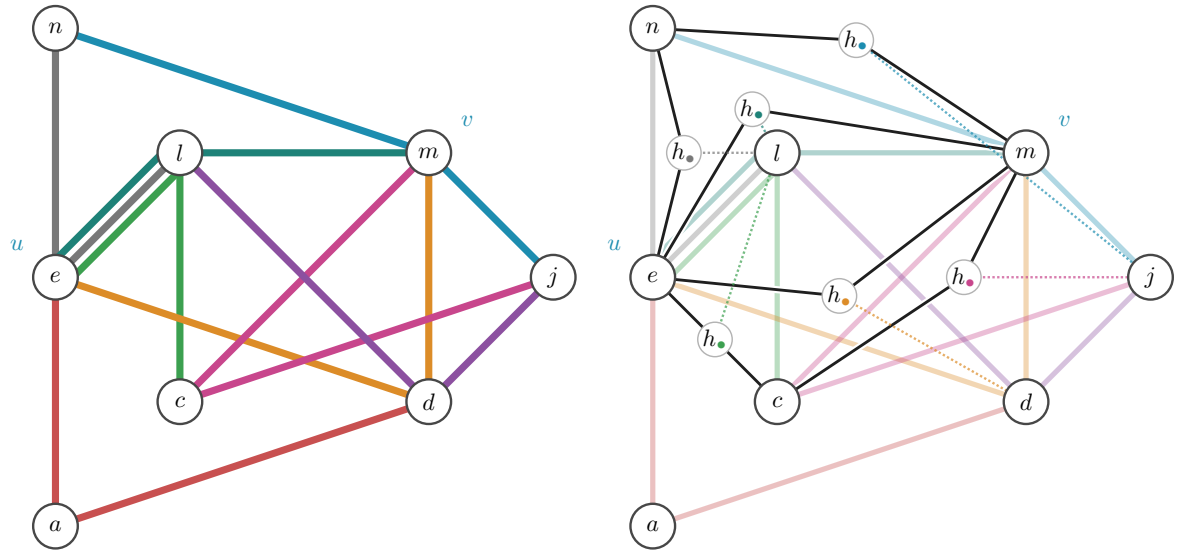


Figure 2.3. (a) An undirected 3-uniform hypergraph. (b) Its gate network for the pair u, v. The four black routes use four different gates, so $\lambda(u, v) = 4$. The vertex v lies in only four hyperedges, so these four gates form a cut and no fifth route exists. Each gate is drawn as one point. Dotted lines join it to the third vertex of its hyperedge.

power (number of entries). For graphs the number of entries grows like n^2, and for r-uniform hypergraphs like n^r. A larger m gives more values per entry. Direction gives more entries, since both directions of a pair are chosen separately. Directed hypergraphs have the most candidates, because each r-set carries r possible hyperedges, one for each tail. The separation does not change the candidates. It only changes the test for each candidate.

So there are $2^{\binom{n}{2}}$ simple graphs and $2^{n(n-1)}$ simple digraphs on n labelled vertices. With multiplicities at most $m-1$ there are $m^{\binom{n}{2}}$ multigraphs and $m^{n(n-1)}$ directed multigraphs. Figure 2.5 shows how fast this grows.

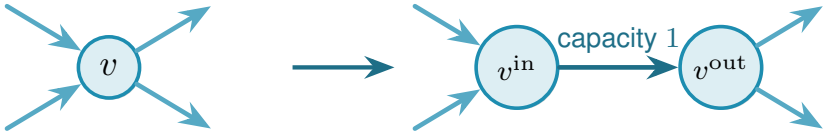
2.3.1 Pruned search

A pruned search skips whole families of graphs. Fix an order on the possible edges and decide them one at a time. A partly built graph stands for all graphs that agree with it so far. A pruning rule can discard this whole family.

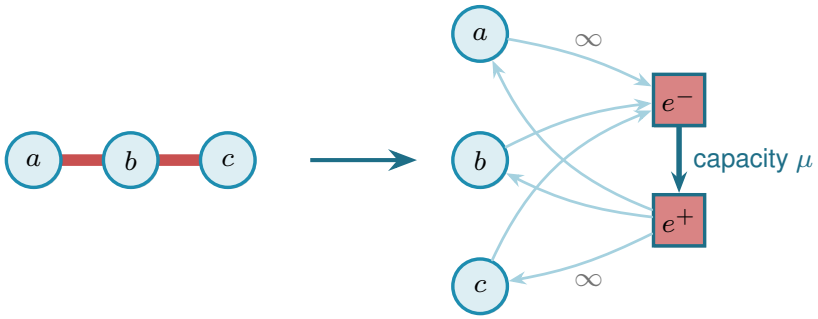
The first rule uses that adding an edge never lowers connectivity. If a partly built graph already breaks the condition, so does every way to complete it, and the branch is dropped.

The second rule bounds the size of every completion: the edges already placed, plus the number of undecided edges times the largest allowed multiplicity. If this cannot beat the best graph found so far, the branch is dropped (Figure 2.6).

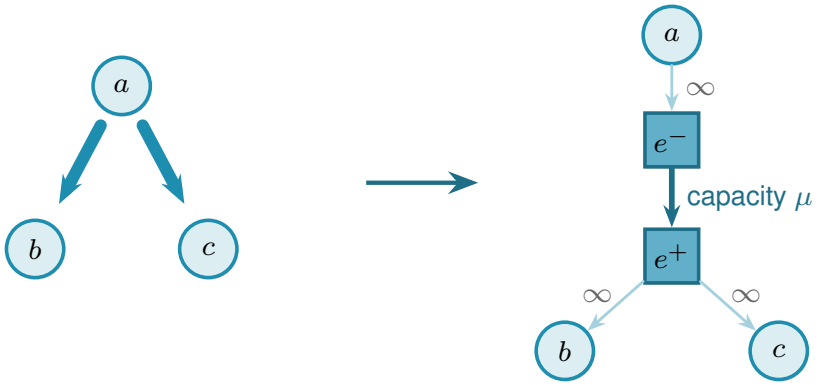
Both rules only drop graphs that are infeasible or cannot beat the best one. So a finished pruned search gives the same answer as the full search.



(a) Splitting each vertex v into v^{in} and v^{out}, joined by an arc of capacity one, turns vertex-disjoint routes into arc-disjoint routes. Only this inner arc has capacity one.



(b) Transform a hyperedge into a gate with capacity equal to the multiplicity of the hyperedge.



(c) The directed hyperedge is transformed into a gate with one tail and two heads. This gate has capacity equal to the multiplicity of the hyperedge and every route must enter at the tail and leave at a head.

Figure 2.4. The building blocks of the connectivity checker: vertex splitting, hyperedge gates and directed hyperedge gates.

2.3.2 Generating search

Many graphs in the full search are isomorphic. Another method avoids this. Prof. Jan Goedgebeur pointed us to the `nauty` toolkit of McKay and Piperno [13]. Its program `geng` lists the undirected graphs on n vertices, one from each isomorphism class. For directed multigraphs, our program uses each such graph as a *support*. Each support edge $\{u, v\}$ gets multiplicities $\mu(u, v)$ and $\mu(v, u)$ between 0 and $m-1$, not both zero. Different supports give non-isomorphic digraphs. After each choice the checker tests the partial digraph, and a branch that breaks the condition is dropped. Digraphs that reach the target size are put in a canonical form, so each isomorphism class is reported once. Different supports can be handled in parallel. We wrote our own step for choosing the multiplicities, because it can prune after each choice.

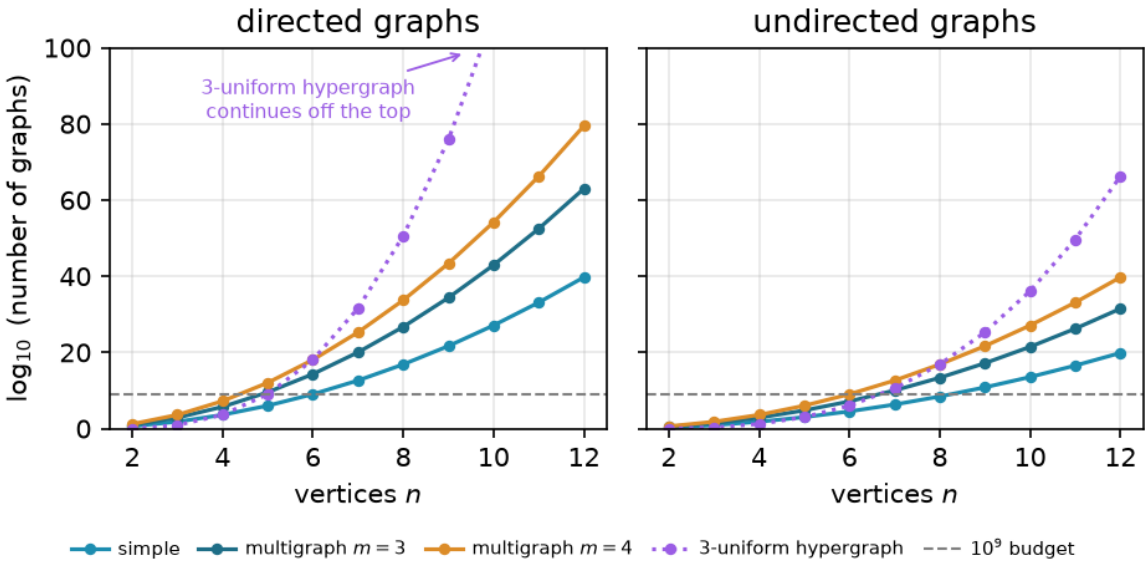


Figure 2.5. The number of candidates for blind enumeration, on a logarithmic scale. Every model passes 10^9 candidates before $n = 10$. Directed models grow fastest, because they have more entries to choose.

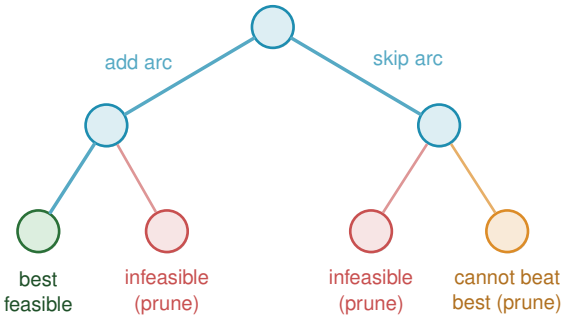


Figure 2.6. Each level of the partially built graph fixes one adjacency. Infeasible partial graphs (red) and branches unable to beat the incumbent (orange) are discarded.

2.4 The heuristic search

For larger n the full search is impossible. Instead we search for dense graphs in a guided way. A *heuristic* is a rule that moves the search towards better graphs. A *metaheuristic* also allows moves that make the graph worse, so that the search does not get stuck.

For graphs and digraphs, each candidate gets an *energy*

$$\Phi(G) = -|E(G)| + \Lambda \cdot \max\bigl(0,\ c^{\max}(G) - (m-1)\bigr).$$

Here $c^{\max}(G)$ is $\lambda^{\max}(G)$ or $\kappa^{\max}(G)$, depending on the separation, and $\Lambda > 0$ is a fixed penalty. Low energy means many edges and no excess connectivity. The search starts from the empty graph and changes one multiplicity by one at each step. We use *tabu search* [14]. At each step it makes the best allowed move, even if every move raises the energy. After a move, the changed pair is *tabu* for a fixed number of steps, so the search cannot undo the move at once. A move

that gives a better graph than any seen so far is always allowed. When the search stalls, it makes a small change to escape.

The program also implements simulated annealing [15] on the same energy. It accepts a worse move with a probability that decreases over time [16].

Every feasible graph can be built from the empty graph by adding edges one at a time, staying feasible. Still, it can help to pass through infeasible graphs: exchanging two edges may need an extra edge first. Tabu search allows this, but it only keeps feasible results.

The search is restarted several times with different random choices, and the densest feasible graph is kept. The restarts are independent and can run in parallel.

2.4.1 Random greedy growth for hypergraphs

For hypergraphs we use random greedy growth. Each run starts with the empty hypergraph and a shuffled list of all possible hyperedges: each hyperedge once in the simple model, and $m - 1$ times in the multi model. The procedure adds the hyperedges one by one, and keeps each one only if the hypergraph stays feasible. This is repeated with new random orders, and the densest result is kept.

At the end of a run no single hyperedge can be added, because a rejected hyperedge stays infeasible when more hyperedges are added. This does not mean the result is as large as possible: an early choice can block several later ones. We use this method instead of tabu search because there are many possible hyperedges, for example $\binom{n}{r}$ in the undirected simple case.

2.4.2 Measuring only near the connectivity cap

Removing an edge never raises $\lambda^{\max}$. Adding an edge raises it by at most one, because at most one route of a largest family uses the new edge. The same holds for $\kappa^{\max}$. So the search does not need to run the checker at every step. Removing an edge from a feasible graph keeps it feasible. Adding an edge to a graph with $\lambda^{\max} \leq m - 2$ also keeps it feasible. The search keeps an upper bound on $\lambda^{\max}$: it adds one for each accepted addition, and it computes the exact value again at fixed intervals. The checker is only called when an addition could break the condition, or when an edge is removed from an infeasible graph.

Chapter 3

Synthesis, Results, and Open Problems

This chapter collects the results of Chapter 1 and the appendix, model by model. It also lists the main open problems.

3.1 The directed frontier

For simple digraphs we know two dense constructions.

Construction 3.1 (Competing dense directed families)**.** Let $n \geq m \geq 2$. The directed hub and the augmented bipartite digraph (Constructions A.13 and A.15) have

$$A_m(n) = m(n-1), \qquad Q_m(n) = \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor .$$

arcs. Both have maximum local arc-connectivity $m-1$. So

$$\ell_m^{\mathrm{dir}}(n) \geq \max\{A_m(n), Q_m(n)\}.$$

For fixed m and large n, the augmented bipartite digraph has more arcs. By Theorem 1.12, the optimum is at most $O_m(n)$ larger. At $m = 2$, Theorem 1.7 gives the exact value, and the bipartite digraph is optimal for $n \geq 7$. Figure 3.1 compares the two counts at $m = 3$.

Section 3.3.5 discusses the asymptotics. It is open whether the arc and vertex values are equal for $m \geq 3$. With parallel arcs allowed, Theorem 1.13 gives the exact value.

3.2 Why direction makes the value quadratic

For undirected graphs the maximum is linear in n (Theorem 1.3). For digraphs it is quadratic. In Construction A.14, all arcs go from one part to the other, so a directed route can cross between the parts only once. This digraph has $\lfloor n^2/4 \rfloor$ arcs and $\lambda^{\max} = 1$. In contrast, an undirected graph with $\lambda^{\max} \leq 1$ has no cycle, so it has at most $n-1$ edges.

Construction A.15 extends this idea to every m (Figure 3.2). Each vertex of the larger part gets $m-2$ extra arcs from other vertices of that part. These arcs give $m-2$ short detours. They raise $\lambda^{\max}$ to $m-1$ and the number of arcs to $\lfloor (n+m-2)^2/4 \rfloor$. At $m = 3$ and $n = 10$, this digraph has 30 arcs, while the hub has 27.

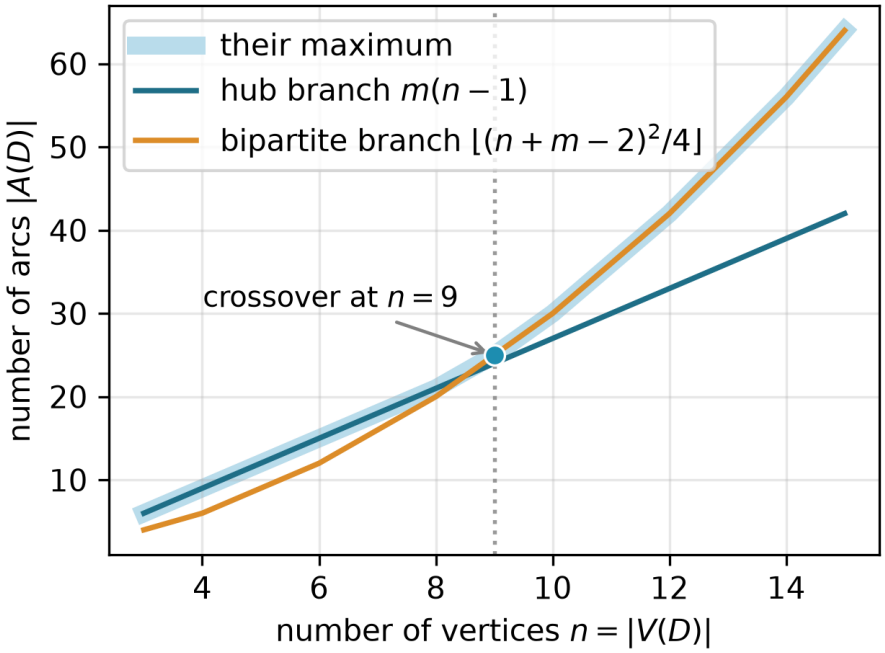


Figure 3.1. The arc counts of the hub and of the augmented bipartite digraph at $m = 3$, and their maximum. These are lower bounds only.

An upper bound must explain why no digraph has many more arcs than this construction. Theorem 1.12 shows that the optimum is at most $O_m(n)$ larger. The linear coefficient is not known. Proposition A.26 also describes the shape: after deleting $O(\sqrt{m}\, n^{3/2})$ arcs, every feasible digraph has all its arcs from one part to another.

3.3 Current status of the sixteen variants

Each of the eight models has an edge problem and a vertex problem. Table 3.1 gives an overview, and the subsections below explain it.

3.3.1 Simple undirected graphs

For edge separation, the answer is known for all n:

$$\ell_m(n) = \begin{cases} \binom{n}{2}, & n < m, \\ \lfloor m(n-1)/2 \rfloor, & n \geq m. \end{cases}$$

For $n < m$ the complete graph is feasible. For $n \geq m$, the hub-and-spokes graph attains Mader's bound. The Gomory–Hu proof shows why simplicity matters: at most $n-1$ edges cross only one tree edge, and every other edge crosses at least two (Theorems 1.3 and A.8).

For vertex separation, the edge formula is still exact for $m \leq 4$. At $m = 5$ the two values agree up to $n = 13$, but from $n = 14$ the vertex value is $\lfloor 8n/3 \rfloor - 4$. The first difference is 33 against 32 at $n = 14$ (Theorems 1.5 and 1.6).

For $m \geq 6$ the vertex value is open. Theorem A.11 reduces it to a knapsack problem over 2-

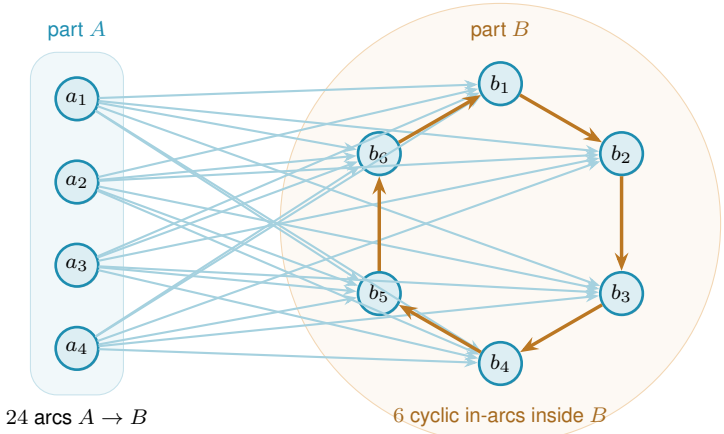


Figure 3.2. The augmented bipartite digraph of Construction A.15 at $m = 3$, $n = 10$. All arcs between the parts go from A to B. Inside B, each vertex gets one arc from the previous vertex on a cycle. A pair a_i, b_j has the direct arc and one detour, so $\lambda^{\max} = 2$. The digraph has $24 + 6 = 30$ arcs, against 27 for the hub.

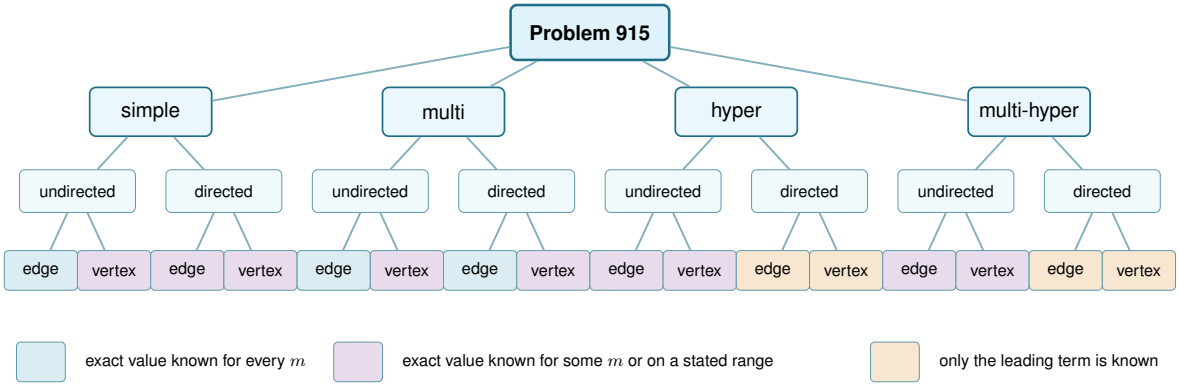


Figure 3.3. Overview of the sixteen variants. Blue: the exact value is known for every m. Purple: the exact value is known for some m or on a stated range. Amber: only the leading term is known.

connected blocks, with a limiting rate c_m. Known results give $c_m = m/2$ for $m \le 4$ and $c_5 = 8/3$ (Theorem 1.6). For $m \ge 6$ they give $m/2 < c_m \le 2(m-1)$.

3.3.2 Undirected multigraphs

For edge separation the value is $L_m(n) = (m-1)(n-1)$, attained by a tree with every edge repeated $m-1$ times (Theorem 1.4).

For vertex separation, parallel edges count as separate routes, so the problem differs from the simple one. Lemma A.43 splits $\kappa(u, v)$ into the parallel edges and the longer routes. At $m = 2$ this forces a forest without repeated edges, so $K_2(n) = n - 1$. At $m = 3$, the incidence-rank bound and a doubled tree give $K_3(n) = 2(n-1)$ (Theorem 1.11 and Proposition A.40). At $m = 5$ the thickened theta gives $K_5(4) \ge 14 > 12 = L_5(4)$ (Section A.10).

The block reduction shows how a good small block can be repeated. Thickened cliques and thickened complete bipartite graphs give bounds of order $m^2 n$ as $m \to \infty$ and $n/m \to \infty$. The constants are $3 - 2\sqrt{2}$ below and 2 above. The true constant is not known.

Model	Edge or arc separation	Vertex separation
Simple graph	Exact edge formula established	Classical exact values for $m \leq 5$, open for $m \geq 6$
Multigraph	$(m-1)(n-1)$ established	$m = 2, 3$ established, with a general block reduction
Simple hypergraph	Cut bound established, attainment on a stated range	$m = 2$ established, $m = 3$ bound established with stated attainment, higher thresholds open
Multihypergraph	Same established cut bound, with additional repeated-edge constructions	$m = 2$ established, $m = 3$ bound and stated attainment established
Simple digraph	Exact $m = 2$ value established, $n^2/4 + \Theta_m(n)$ established for $m \geq 3$	Same asymptotic order, with equality to the arc value open for $m \geq 3$
Directed multigraph	Exact arc formula $(m-1)M(n)$ established	Established leading term $(m-1)n^2/4$, larger than the arc value for some small n
Directed simple hypergraph	Leading term $(m-1)n^2/[4(r-1)]$ established	Same established leading term
Directed multihypergraph	Same established leading term	Same established leading term

Table 3.1. Status of the sixteen problems. Here $m \geq 2$, hypergraphs have $r \geq 3$, and asymptotic statements are for fixed m and r as $n \to \infty$.

3.3.3 Simple undirected hypergraphs

In an r-uniform hypergraph one hyperedge joins several vertices, but it can carry only one route of an edge-disjoint family. The edge bound is

$$\ell_m^{(r)}(n) \leq \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor .$$

It is attained when $m - 1 \leq \binom{n-2}{r-2}$, that is, when there are enough different hyperedges (Theorems 1.9 and A.35). This condition matters: at $n = r = m = 3$ there is only one possible hyperedge, while the bound is two.

For vertex separation, $k_2^{(r)}(n) = \lfloor (n-1)/(r-1) \rfloor$. Here feasibility means that the incidence graph is a forest, and a star hypertree attains the bound. At $m = 3$ the incidence-rank bound gives $\lfloor 2(n-1)/(r-1) \rfloor$, attained by simple hypergraphs for $2 < r < n$. For $m \geq 4$ the vertex problem is open. The edge bound does not help here, because the vertex condition is weaker.

3.3.4 Undirected multihypergraphs

The edge bound also holds when hyperedges repeat. A star hypertree with every hyperedge repeated $m-1$ times attains $(m-1)(n-1)/(r-1)$ when $r-1$ divides $n-1$. At $n = r = m = 3$, two copies of the only hyperedge reach the bound, which no simple hypergraph can do. Repeated hyperedges can also reach the bound in other cases: the construction with twelve hyperedges at $(n, m, r) = (6, 6, 3)$ meets the bound 12 (Construction A.37).

For vertex separation, repeated hyperedges do not help at $m = 2$, because two copies of a

hyperedge already give two separated routes. So the value is again $\lfloor (n-1)/(r-1) \rfloor$. At $m = 3$ the upper bound is the same as for simple hypergraphs. A star hypertree with every hyperedge doubled attains it when $r - 1$ divides $n - 1$ (Proposition A.40). For $m \geq 4$ the vertex problem is open.

3.3.5 Simple directed graphs

In a digraph, the one-way bipartite digraph already has $\lfloor n^2/4 \rfloor$ arcs with $\lambda^{\max} = 1$. At $m = 2$ the value under both separations is $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$. The proof uses the reachability skeleton.

For fixed $m \geq 3$, both values are $n^2/4 + \Theta_m(n)$. The upper bound also follows from a theorem of Huang and Lyu [12, Theorem 2] with $t = m - 1$. A feasible digraph has at most $m - 1$ paths of length two between any two vertices, since paths with different midpoints are internally vertex-disjoint. Their theorem then gives $|A| \leq n^2/4 + (m-1)n + O_m(1)$ for large n. The augmented bipartite digraph gives $n^2/4 + (m-2)n/2 + O_m(1)$ from below. Theorem 1.12 gives an elementary proof, with a bound that holds for every n.

The exact value and the linear coefficient are open. It is also open whether the arc and vertex values are equal for $m \geq 3$. An upper bound for the arc problem does not bound the vertex problem, because more digraphs are vertex-feasible.

3.3.6 Directed multigraphs

For arc separation the value is $L_m^{\mathrm{dir}}(n) = (m-1)M(n)$. The proof removes a reachability skeleton $m - 1$ times. A bidirected tree and the one-way bipartite digraph, with every arc repeated $m - 1$ times, attain the two terms. For $n \geq \max(8, 2m)$ the bipartite construction is the only extremal multigraph (Theorem A.32).

For vertex separation the answer can be larger. On three vertices, take every arc with multiplicity 4. Every route of length at least two passes through the third vertex, so $\kappa \leq 4 + 1 = 5$. This digraph has 24 arcs, so $K_6^{\mathrm{dir}}(3) \geq 24$, while $L_6^{\mathrm{dir}}(3) = 20$. For fixed m, the shadow bound at $r = 2$ and the bipartite construction give $K_m^{\mathrm{dir}}(n) = (m-1)n^2/4 + O_m(n)$ (Theorem 1.14 and Corollary A.53). The exact value is open. A block construction gives a lower bound (Proposition A.55).

3.3.7 Simple directed hypergraphs

In the forward model a hyperedge has one tail and $r - 1$ heads. For fixed $r \geq 3$ and $m \geq 2$, the leading term under both separations is $(m-1)n^2/(4(r-1))$. The upper bound uses the shadow digraph. The lower bound uses a two-layer construction: all tails on one side, all heads on the other (Proposition A.56).

For small n there may not be enough different head sets for this construction. Then another construction can be larger: give every vertex the same number of outgoing hyperedges. For example, at $n = m = 6$ and $r = 3$, every vertex gets five outgoing hyperedges, which gives 30 hyperedges. All routes from a vertex start with one of its five outgoing hyperedges, so the

hypergraph is feasible (Construction A.57).

3.3.8 Directed multihypergraphs and orientation choices

With repeated hyperedges, the two-layer construction no longer needs different head sets. The leading term is still $(m-1)n^2/(4(r-1))$ under both separations. At $r=2$ these are the directed multigraphs. For simple digraphs ($r=2$) the leading coefficient is $1/4$ instead, so the simple case needs $r \geq 3$.

We can also orient a hyperedge in other ways. In the *backward* model a hyperedge has one head and $r-1$ tails. Reversing all arcs turns it into the forward model (Lemma A.58). In the *general* model a hyperedge is split in any way into a non-empty set of tails and a non-empty set of heads. All three models have the same leading term (Theorem A.61). It is open whether the forward and general models have the same exact values.

3.4 Scope of the program

The program measures connectivity in all sixteen variants with one max-flow checker. It uses tabu search and simulated annealing for graphs, and random greedy growth for hypergraphs. An exhaustive search gives an exact value only when it covers every candidate or excludes it by a correct pruning rule. A heuristic search only gives lower bounds.

3.5 Open problems

Four directions remain open.

- ★ For directed graphs, determine the exact values, linear corrections and extremal structures, including whether the two separations agree in the simple model.
- ★ For undirected vertex separation, determine the simple value at $m \geq 6$ and the multigraph value at $m \geq 4$, including the best blocks and growth rates.
- ★ For undirected hypergraphs, determine edge attainment outside the range where it is established and find vertex bounds beyond the $m=3$ rank argument.
- ★ For directed hypergraphs, determine finite values and compare the effects of multiplicity, separation and orientation beyond the established leading term.

Three concrete open questions:

- ⛤ Find $\ell_m^{\mathrm{dir}}(n)$ for $m \geq 3$.
- ⛤ Is $k_m^{\mathrm{dir}}(n) = \ell_m^{\mathrm{dir}}(n)$ for $m \geq 3$?
- ⛤ Is $K_4(n) = L_4(n) = 3(n-1)$ for every n?

Appendix A

Proofs

This appendix contains all proofs, and the lemmas they need. The definitions are in Chapter 1. New terms are defined where they are first used.

The appendix has three parts. PART I contains the classical tools and the proof of Mader's theorem. PARTS II and III contain the new results. The three main proof ideas are the reachability skeleton (directed multigraphs), the incidence-rank lemma (hypergraphs at $m = 3$), and the two-step budget (directed graphs and hypergraphs).

Part I. Classical tools and Mader's theorem

The theorems of Menger and Gomory–Hu, cited without proof, and a full proof of Mader's theorem with the Gomory–Hu tree.

A.1 Menger, Gomory–Hu, and the degree bound

We use two classical results.

Theorem A.1 (Menger [5, 6])**.** *Let u and v be distinct vertices of a multigraph G. The largest number of pairwise edge-disjoint u–v paths equals the smallest number of edges whose removal leaves no u–v path. The same holds in a directed multigraph, with directed paths and arcs.*

Theorem A.2 (Gomory–Hu [17])**.** *Let G be a multigraph on the vertex set V. There is a tree T on V, with a weight $c(t)$ on each tree edge t, such that:*

(1) Removing t from T splits V into two parts S and $V \setminus S$, and $c(t)$ is the number of edges of G with one end in S and one end in $V \setminus S$.

(2) For distinct u, v, the value $\lambda_G(u, v)$ is the smallest weight on the u–v path of T.

Such a tree is a *Gomory–Hu tree* of G. Its edges need not be edges of G. If $t = xy$ is a tree edge, then the x–y path of T is t itself, so $c(t) = \lambda_G(x, y)$ by (2). This thesis uses the theorem only for undirected graphs.

We also use a degree bound.

Lemma A.3 (Degree bound)**.** *Every pair u, v of distinct vertices of a graph G satisfies $\lambda_G(u, v) \leq \min(\deg u, \deg v)$, and every ordered pair of a digraph D satisfies $\lambda_D(u, v) \leq \min(d^+(u), d^-(v))$.*

Proof. Every edge-disjoint u–v path uses a distinct edge at u, so there are at most $\deg u$ of them, and likewise at most $\deg v$. In the directed case each arc-disjoint path leaves u by a distinct out-arc and enters v by a distinct in-arc. □

A.2 Mader's theorem in full

Let T be a Gomory–Hu tree of G. For an edge e of G, let $\mathrm{dist}_T(e)$ be the number of tree edges on the path in T between the ends of e (Figure A.1). An edge $e = uv$ of G *crosses* a tree edge t if u and v lie in different parts when t is removed from T. This happens exactly when t lies on the u–v path of T. So e crosses $\mathrm{dist}_T(e)$ tree edges.

The proof has three steps.

(1) The tree weights add up to at most $(m - 1)(n - 1)$.

(2) The same sum counts every edge of G once for each tree edge it crosses (Lemma A.6).

(3) In a simple graph, at most $n-1$ edges cross only one tree edge, and every other edge crosses at least two (Lemma A.5).

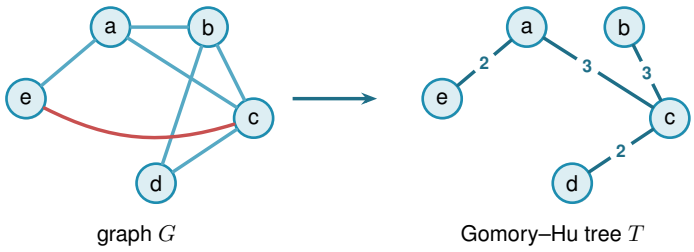


Figure A.1. A graph G and a Gomory–Hu tree T on the same vertices. Each edge e of G crosses $\mathrm{dist}_T(e)$ tree edges. In a simple graph, at most $n-1$ edges have $\mathrm{dist}_T(e) = 1$.

Lemma A.4 (Distance-one edges)**.** *At most $n-1$ edges of a simple graph G satisfy $\mathrm{dist}_T(e) = 1$.*

Proof. An edge e has $\mathrm{dist}_T(e) = 1$ exactly when its ends are adjacent in T. The tree T has $n-1$ edges. Since G is simple, each pair of adjacent vertices of T is joined by at most one edge of G. □

Lemma A.5 (Sum of tree distances)**.** $\displaystyle\sum_{e\in E(G)} \mathrm{dist}_T(e) \;\geq\; 2|E(G)| - (n-1).$

Proof. Partition $E(G)$ into $E_1 = \{e : \mathrm{dist}_T(e) = 1\}$ and $E_{\geq 2} = \{e : \mathrm{dist}_T(e) \geq 2\}$. Then

$$\sum_{e\in E(G)} \mathrm{dist}_T(e) \;\geq\; |E_1| + 2\,|E_{\geq 2}| \;=\; 2|E(G)| - |E_1| \;\geq\; 2|E(G)| - (n-1),$$

using Lemma A.4 in the last step. □

Lemma A.6 (Double-counting identity)**.** *Let G be a multigraph and T a Gomory–Hu tree of G with weights c. Then*

$$\sum_{t\in E(T)} c(t) \;=\; \sum_{e\in E(G)} \mathrm{dist}_T(e).$$

Parallel edges are counted separately in the right-hand sum.

Proof. Count the pairs (t, e) where t is a tree edge, e is an edge of G, and e crosses t. For a fixed t, exactly $c(t)$ edges cross t, by property (1) of the Gomory–Hu theorem. For a fixed e, exactly $\mathrm{dist}_T(e)$ tree edges are crossed by e. So both sums count the same pairs.

□

Proof of Theorem 1.3, upper bound. Let G be a simple graph on n vertices with $\lambda_G^{\max} \leq m-1$, and let T be a Gomory–Hu tree of G. Each tree edge $t = xy$ has weight $c(t) = \lambda_G(x, y) \leq m-1$. There are $n-1$ tree edges, so $\sum_t c(t) \leq (m-1)(n-1)$. With Lemmas A.5 and A.6 this gives

$$2|E(G)| - (n-1) \;\leq\; \sum_{e\in E(G)} \mathrm{dist}_T(e) \;=\; \sum_{t\in E(T)} c(t) \;\leq\; (m-1)(n-1).$$

So $2|E(G)| \le m(n-1)$. Since $|E(G)|$ is an integer, $|E(G)| \le \left\lfloor \frac{m(n-1)}{2} \right\rfloor$. □

The factor $\frac{1}{2}$ comes from step (3). In a multigraph step (3) fails, and the bound is $(m-1)(n-1)$ (Theorem 1.4).

Proof of Theorem 1.4. For the lower bound, take any tree on the n vertices and give every edge multiplicity $m-1$. This multigraph has $(m-1)(n-1)$ edges. Two vertices u, v are joined by exactly one path in the tree. Removing the $m-1$ copies of one edge of this path separates u from v. So $\lambda(u,v) \le m-1$ for every pair.

For the upper bound, let $\lambda_G^{\max} \le m-1$ and let T be a Gomory–Hu tree of G. Every edge of G crosses at least one tree edge. So by Lemma A.6, $|E(G)| \le \sum_e \operatorname{dist}_T(e) = \sum_t c(t) \le (m-1)(n-1)$. □

Construction A.7 (Hub-and-spokes graph)**.** Let $n \ge m \ge 2$. The graph $H_{m,n}$ has a vertex h and further vertices $v_1, \dots, v_{n-1}$. Join h to every v_i. On $v_1, \dots, v_{n-1}$, add a graph with $\lfloor (m-2)(n-1)/2 \rfloor$ edges in which every vertex has degree at most $m-2$. For $n \ge 3$ such a graph exists by Lemma A.38 with uniformity 2. If $m-1$ divides $n-1$, one can take disjoint copies of K_{m-1}. Then $H_{m,n}$ consists of copies of K_m that share the vertex h.

Theorem A.8 (Extremal graphs exist)**.** *For all $n \ge m \ge 2$ the graph $H_{m,n}$ has $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ edges and $\lambda^{\max} \le m-1$.*

Proof. The number of edges is $(n-1) + \left\lfloor \frac{(m-2)(n-1)}{2} \right\rfloor = \left\lfloor \frac{m(n-1)}{2} \right\rfloor$. Of any two vertices, at least one is some v_i. This vertex has degree at most $1 + (m-2) = m-1$. So Lemma A.3 gives $\lambda(u,v) \le m-1$. □

So for $n \ge m$ the bound of Theorem 1.3 is attained. The next theorem describes what equality means for the Gomory–Hu tree.

Theorem A.9 (Equality in the Gomory–Hu bound)**.** *Let G be simple with $\lambda_G^{\max} \le m-1$ and $|E(G)| = \left\lfloor \frac{m(n-1)}{2} \right\rfloor$, and let T be a Gomory–Hu tree of G.*

(a) If $m(n-1)$ is even, then every tree edge has weight $m-1$, every tree edge is an edge of G, and every edge of G has $\operatorname{dist}_T(e) \le 2$.

(b) If $m(n-1)$ is odd, then exactly one of these three statements fails, and it fails by one: one tree edge has weight $m-2$, or one tree edge is not an edge of G, or one edge of G has $\operatorname{dist}_T(e) = 3$.

Proof. Each inequality in the proof of Theorem 1.3 has a non-negative integer gap:

$$S_1 = (m-1)(n-1) - \sum_t c(t), \qquad S_2 = \sum_{e:\, \operatorname{dist}_T(e) \ge 2} \big(\operatorname{dist}_T(e) - 2\big), \qquad S_3 = (n-1) - |E_1|.$$

Here S_1 counts tree weights below $m-1$, S_2 counts how far edges go beyond tree distance two, and S_3 counts tree edges that are not edges of G. The proof gives two expressions for the same sum:

$$\sum_e \mathrm{dist}_T(e) \;=\; 2|E| - |E_1| + S_2 \qquad \text{and} \qquad \sum_e \mathrm{dist}_T(e) \;=\; \sum_t c(t) \;=\; (m-1)(n-1) - S_1.$$

We substitute $|E_1| = (n-1) - S_3$ into the first and compare with the second:

$$\begin{aligned} 2|E| - (n-1) + S_2 + S_3 \;&=\; (m-1)(n-1) - S_1, \\ \text{so} \qquad 2|E| \;&=\; m(n-1) - (S_1 + S_2 + S_3). \end{aligned}$$

If $m(n-1)$ is even, then $|E| = m(n-1)/2$ forces $S_1 = S_2 = S_3 = 0$. If $m(n-1)$ is odd, then $|E| = (m(n-1)-1)/2$ gives $S_1 + S_2 + S_3 = 1$, so exactly one gap equals one. □

For $m = 2$ this shows that the extremal graphs are exactly the trees on the n vertices. Indeed, every tree edge then has weight one and is an edge of G, so G contains T and has only $n-1$ edges.

Remark A.10 (Forcing and avoiding counts). This thesis maximises the number of edges with no m disjoint routes between any pair. Sørensen and Thomassen, and the Erdős Problems database, give instead the least number of edges that forces such a pair [3, 11]. Their number is one more than ours.

For example, Theorem 4 of [11] gives the forcing number $\lfloor 8n/3 \rfloor - 3$ for $n \geq 6$, except 16 at $n = 7$ and 28 at $n = 12$. Subtracting one gives our values. For $6 \leq n \leq 13$ the paper also gives $\lfloor 5(n-1)/2 \rfloor + 1$. This gives the two cases of Theorem 1.6. At $n = 13$ both maxima are 30, and the edge and vertex values first differ at $n = 14$.

The same care is needed for the Bollobás–Erdős conjecture of Section A.2.1. Its forcing number $1 + \binom{m}{2}p$ is for graphs on $1 + (m-1)p$ vertices. Each copy of K_m adds $\binom{m}{2}$ edges but only $m-1$ vertices, so in our notation the rate is $m/2$ edges per vertex.

A.2.1 The vertex problem is a block problem

The vertex problem $k_m(n)$ is open for $m \geq 6$. This section reduces it to 2-connected graphs (Theorem A.11).

A graph on more than k vertices is *k-connected* if removing any $k-1$ vertices leaves it connected. A *cut vertex* is a vertex whose removal disconnects the graph. A *bridge* is an edge whose removal disconnects its component. A *block* is a maximal connected subgraph that has an edge and no cut vertex of its own. In a simple graph, a block is either a bridge or a maximal 2-connected subgraph.

Below, a *knapsack* is the following problem: choose blocks of sizes b_i, each costing $b_i - 1$ vertices and giving $h_m(b_i)$ edges, with total cost at most $n-1$.

Theorem A.11 (The vertex problem is a knapsack over blocks)**.** *For $b \geq 2$ let*

$$h_m(b) \;:=\; \max\bigl\{\, |E(B)| \;:\; B \text{ 2-connected or a single edge},\ |V(B)| = b,\ \kappa_B^{\max} \leq m-1 \,\bigr\},$$

with $h_m(b) := -\infty$ when no such B exists. Then for all $m \geq 2$ and $n \geq 2$,

$$k_m(n) \;=\; \max\Bigl\{ \sum_i h_m(b_i) \;:\; b_i \geq 2,\ \sum_i (b_i - 1) \leq n-1 \Bigr\}.$$

Consequently $k_m(n) \geq k_m(n_1) + k_m(n_2)$ whenever $n = n_1 + n_2 - 1$, and

$$c_m \;:=\; \lim_{n\to\infty} \frac{k_m(n)}{n-1} \;=\; \sup_{b\geq 2} \frac{h_m(b)}{b-1}$$

exists, the limit being approached from below. The supremum is finite: Remark A.12 gives $k_m(n) < 2(m-1)n$, so $c_m \leq 2(m-1)$.

Proof. Let G be feasible, and let u, v lie in a common block B. No u–v path leaves B: it would have to leave and come back through the same cut vertex, and a path visits each vertex once. So $\kappa_G(u,v) = \kappa_B(u,v)$. Every block is a subgraph of G, so it is feasible too. Every edge lies in exactly one block, so $|E(G)| = \sum_B |E(B)| \leq \sum_B h_m(|V(B)|)$. In each component, $\sum_B (|V(B)| - 1)$ is the number of vertices minus one. So over all of G, $\sum_B (|V(B)| - 1) \leq n-1$. This proves $\leq$.

For $\geq$, take blocks $B_1, \ldots, B_t$ with $\sum (b_i - 1) \leq n-1$. Glue them at one common vertex, and keep them disjoint otherwise. This graph has $1 + \sum (b_i - 1) \leq n$ vertices. Add isolated vertices to reach n. Its blocks are the B_i, so pairs inside one block keep their value of κ. Two vertices in different blocks are separated by the common vertex, so $\kappa \leq 1 \leq m-1$. So the graph is feasible, with $\sum_i |E(B_i)|$ edges.

Superadditivity is the special case of gluing two extremal graphs at a single vertex. Write $a_n := k_m(n)$ and $b_t := a_{t+1}$ for $t \geq 1$. The gluing inequality gives $b_{s+t} \geq b_s + b_t$ for $s, t \geq 1$. Fekete's lemma [18] therefore gives $b_t/t \to \sup_{t\geq 1} b_t/t$, or equivalently $a_n/(n-1) \to \sup_{n\geq 2} a_n/(n-1)$. Also $a_n/(n-1) \geq h_m(b)/(b-1)$ whenever $(b-1) \mid (n-1)$, while $a_n \leq \sup_b h_m(b)/(b-1) \cdot (n-1)$ from the display. Hence the limit is $\sup_b h_m(b)/(b-1)$. □

We note three consequences.

First, at $m = 3$ the reduction gives Theorem 1.5 again. A block with $\kappa^{\max} \leq 2$ is an edge or a cycle. Otherwise it contains a *theta*: three internally disjoint paths with the same two ends, which break the condition. So $h_3(2) = 1$ and $h_3(b) = b$ for $b \geq 3$. The rate $b/(b-1)$ is largest at $b = 3$, so the knapsack uses triangles: $k_3(n) = 3\lfloor (n-1)/2 \rfloor + ((n-1) \bmod 2) = \left\lfloor \frac{3(n-1)}{2} \right\rfloor$.

Second, at $m = 4$, Theorem 1.5 gives $h_4(b) = 2(b-1)$ for every $b \geq 4$. So the rate $h_4(b)/(b-1)$ is 2 for $b \geq 4$, against 1 at $b = 2$ and $3/2$ at $b = 3$. The knapsack uses blocks of size at least four, and $k_4(n) = 2(n-1)$ for $n \geq 4$. This uses Theorem 1.5, so it is not a new proof. It does show the extremal block: the hub-and-spokes graph $H_{4,b}$ of Section A.2, which is a hub

joined to all vertices of a cycle. It is a single block with $2(b-1)$ edges.

Third, the Bollobás–Erdős construction, copies of K_m sharing one vertex, is the case $b = m$: $h_m(m) = \binom{m}{2}$, with rate $m/2$. Their conjecture says that $c_m = m/2$, so that no block beats K_m. A *bouquet* is a set of blocks that share one vertex and are otherwise disjoint. Its rate is a weighted average of the rates of its blocks. So a counterexample must contain one 2-connected block with rate above $m/2$.

The conjecture is false. Sørensen and Thomassen [11] showed that for each fixed m,

$$c_m \;\geq\; \frac{m(m-1)-2}{2m-3} \;=\; \frac{m}{2} + \frac{1}{4} + O(1/m).$$

This is larger than $m/2$ for every $m \geq 5$, because

$$2\big(m(m-1)-2\big) - m(2m-3) \;=\; m-4.$$

A *separating pair* is a pair of vertices whose removal disconnects the graph. A graph is 3-*connected* if removing any two vertices leaves it connected.

Their example is built step by step. Start with $G^0 = K_m$ minus an edge. To build G^j, take G^{j-1} and two new copies of G^0, and glue them in a cycle: each one shares one vertex with the next. Each step adds $2m-3$ vertices and $m(m-1)-2$ edges, which gives the rate above. Gluing in a cycle creates no cut vertex, so G^j is a single block. So Theorem A.11 sees G^j as one block of size $m + (2m-3)j$ and does not help to find it. The graph does have separating pairs, so a decomposition along separating pairs [19, 20] would see its structure. The recursion beats the rate $m/2$ only when $j > 2/(m-4)$. So the smallest member of this family that beats copies of K_m has 26 vertices at $m = 5$, 24 at $m = 6$ and 18 at $m = 7$. Smaller examples exist. At $m = 5$, Theorem 1.6 gives 33 edges on 14 vertices, and by the bouquet argument above such a graph contains a block with rate above $5/2$.

Remark A.12 (A degeneracy question, and what it would give). Mader's density theorem [21], in the form of [4, Theorem 1.4.3], gives $k_m(n) < 2(m-1)n$: a denser graph has an m-connected subgraph. Feasibility gives more. For example, two adjacent vertices have at most $m-2$ common neighbours, since each common neighbour gives a route of length two next to the edge.

We ask whether every feasible graph has a vertex of degree at most $m-1$. Feasibility passes to induced subgraphs. So a positive answer would mean that every non-empty induced subgraph has a vertex of degree at most $m-1$. Removing such vertices one by one would give

$$k_m(n) \;\leq\; \sum_{i=1}^{n} \min(i-1,\, m-1) \;=\; (m-1)n - \binom{m}{2} \qquad (n \geq m-1),$$

half the constant of Mader's bound.

The answer is yes for $m = 3$. By a theorem of Dirac [22], a graph of minimum degree at least three contains a *subdivision* of K_4: a copy of K_4 in which edges are replaced by internally disjoint paths. Two of its four main vertices are joined by three internally disjoint paths, which breaks the

condition at $m = 3$. For $m \geq 4$ the question is open.

Part II. Directed graphs

The directed constructions, the reachability skeleton, the value at $m = 2$, the bounds for $m \geq 3$, and the exact value for directed multigraphs.

A.3 The directed constructions

Three constructions give the directed lower bounds. Figure 1.7 shows the first two. Here we define them and check their connectivity.

Construction A.13 (Directed hub)**.** For $n \geq m \geq 2$, take a hub h and label the other $N = n - 1$ vertices $0, \ldots, N - 1$. Add the $2N$ hub arcs $h \to i$ and $i \to h$, and on the non-hub vertices the $(m - 2)N$ ring arcs $i \to i + j \bmod N$ for $j = 1, \ldots, m - 2$. The total is $m(n - 1)$ arcs and every non-hub vertex has $d^+ = d^- = m - 1$.

Every ordered pair contains a vertex other than the hub, and that vertex has out-degree and in-degree $m - 1$. So the degree bound gives $\lambda^{\max} \leq m - 1$. At $m = 2$ there are no ring arcs, and we get the *bidirected star*.

Construction A.14 (One-directional bipartite digraph)**.** Split the vertices into parts A and B of sizes $\lfloor n/2 \rfloor$ and $\lceil n/2 \rceil$, and include every arc from A to B and none the other way. The total is $\lfloor n^2/4 \rfloor$ arcs.

No arc leaves B and no arc enters A. So every route has length one, and $\lambda^{\max} = 1$.

Construction A.15 (Augmented bipartite digraph)**.** For $n \geq m \geq 2$, set $|B| = \lceil (n + m - 2)/2 \rceil$ and $|A| = n - |B|$, include every arc from A to B, and inside B give each b_j the $m - 2$ in-arcs from its cyclic predecessors $b_{j-1}, \ldots, b_{j-(m-2)}$. The total is $\lfloor (n + m - 2)^2/4 \rfloor$ arcs.

Lemma A.20 shows that $\lambda^{\max} = m - 1$. For example, take $m = 3$ and $n = 10$. Then $|B| = 6$ and $|A| = 4$. There are 24 arcs from A to B, and 6 arcs forming a cycle inside B. Each pair a_i, b_j has the direct arc and one detour, so $\lambda^{\max} = 2$. This gives 30 arcs, more than the $\max(m(n - 1), \lfloor n^2/4 \rfloor) = 27$ of the first two constructions. The gain comes from making B larger than half of the vertices.

A.4 The reachability skeleton

The next lemma is used for both exact directed results. Section A.5 uses it once, and Section A.7 uses it $m - 1$ times.

Definition A.16 (Reachability skeleton)**.** Let D be a directed multigraph. A *reachability skeleton* of D is a subdigraph R on the same vertices, with at most one copy of each arc, such that for all $u \neq v$: u can reach v in R exactly when u can reach v in D.

A skeleton keeps the information about who can reach whom. It does not keep the number of disjoint paths.

Lemma A.17 (A sparse reachability skeleton always exists)**.** *Every loopless directed multigraph D on $n \geq 2$ vertices has a reachability skeleton R with $|A(R)| \leq M(n)$.*

Proof. We build R in three steps: first inside each strongly connected component, then between components, and then we count. A *strongly connected component* is a maximal set of vertices in which any two can reach each other. Every digraph splits into such components. Figure A.2 shows an example.

Step 1: inside each component. Take a component with n_i vertices and choose a vertex r in it. Since r can reach every vertex of the component, there is an *out-tree*: $n_i - 1$ arcs, all pointing away from r, along which r reaches every vertex. (Grow it one arc at a time.) In the same way there is an *in-tree* of $n_i - 1$ arcs, along which every vertex reaches r. Any two vertices u, v of the component are then joined by a walk $u \to r \to v$ in these two trees, which contains a path. So at most $2(n_i - 1)$ arcs keep all reachability inside the component.

Step 2: between components. Replace each component by one point. Put an arc from one point to another if D has an arc from the first component to the second. This gives a digraph Q_0 on q points, one per component. Q_0 has no directed cycle, since a cycle would merge components. Now delete arcs from Q_0 one at a time, as long as this does not change who can reach whom. Call the result Q.

The graph Q, with directions ignored, has no triangle. Indeed, suppose three points x, y, z of Q are joined in pairs. Since Q has no directed cycle, the three arcs must look like $x \to y$, $y \to z$ and $x \to z$, after renaming. But then $x \to z$ could be deleted, since x still reaches z through y. This contradicts the choice of Q. By Mantel's theorem [23, 24], a triangle-free graph on q vertices has at most $\lfloor q^2/4 \rfloor$ edges. So

$$|A(Q)| \;\leq\; \left\lfloor \frac{q^2}{4} \right\rfloor .$$

Step 3: counting. Let R consist of the two trees of Step 1 in every component, and, for each arc of Q, one arc of D between the corresponding components. By Step 1, reachability inside each component is kept. By Step 2, reachability between components is kept: a route can follow the arcs of Q and move inside each component along the trees. Since $R \subseteq D$, R creates no new reachability. So R is a reachability skeleton. If the components have sizes $n_1, \ldots, n_q$, then

$$|A(R)| \;\leq\; \sum_{i=1}^{q} 2(n_i - 1) \;+\; |A(Q)| \;\leq\; 2(n-q) \;+\; \left\lfloor \frac{q^2}{4} \right\rfloor \;=:\; f(q).$$

It remains to bound $f(q)$ for $1 \leq q \leq n$. Since $\lfloor (q+1)^2/4 \rfloor - \lfloor q^2/4 \rfloor = \lfloor (q+1)/2 \rfloor$ (check

even and odd q),

$$f(q+1) - f(q) \;=\; -2 \;+\; \left\lfloor \frac{q+1}{2} \right\rfloor .$$

These differences never decrease, so f is convex. A convex function on $\{1, \ldots, n\}$ is largest at $q = 1$ or at $q = n$. Since $f(1) = 2(n-1)$ and $f(n) = \lfloor n^2/4 \rfloor$, we get $f(q) \le M(n)$. □

The digraph Q is called the *transitive reduction* of Q_0.

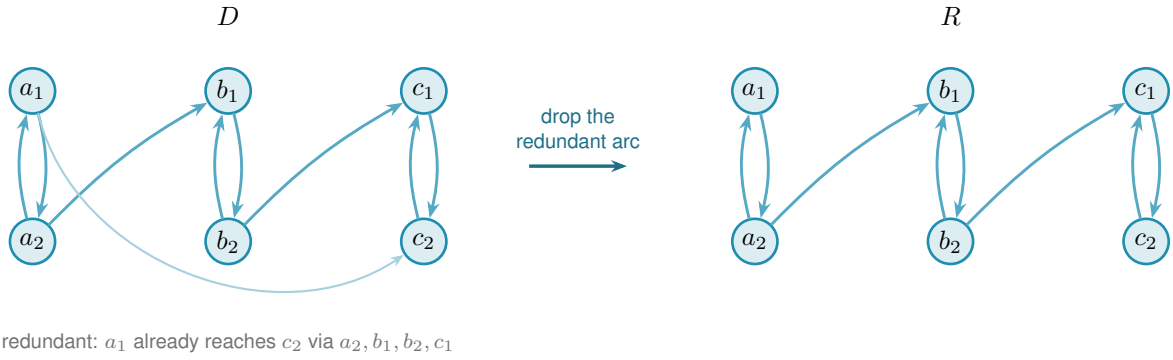


Figure A.2. The construction of Lemma A.17. Each bidirected pair A, B, C is a strongly connected component. Contracting the pairs gives $A \to B \to C$ and the redundant arc $A \to C$. Delete that arc to obtain the triangle-free transitive reduction. The skeleton R keeps the internal arcs of each pair and the two remaining connections between pairs. Although $a_1 \to c_2$ is removed, a_1 still reaches c_2 through B.

A.5 The directed arc value at $m = 2$

This section proves the two values at $m = 2$. We first prove the vertex version with Lemma A.17. The arc version follows, because every arc-feasible digraph is vertex-feasible.

Deleting a vertex cannot raise $\lambda^{\max}$ or $\kappa^{\max}$, because it only removes routes. We use this in Section A.6.1.

Since $\kappa \le \lambda$, every arc-feasible digraph is vertex-feasible. So $k_m^{\mathrm{dir}}(n) \ge \ell_m^{\mathrm{dir}}(n)$. An upper bound for the arc problem gives no upper bound for the vertex problem.

Proof of Theorem 1.8. Write $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$.

Lower bound. The bidirected star (Construction A.13 at $m = 2$) and the one-way bipartite digraph (Construction A.14) have $\lambda^{\max} = 1$. Since $\kappa \le \lambda$, they also have $\kappa^{\max} \le 1$. So $\ell_2^{\mathrm{dir}}(n) \ge M(n)$ and $k_2^{\mathrm{dir}}(n) \ge M(n)$.

Upper bound. Let D be a simple digraph on $n \ge 2$ vertices with $\kappa_D^{\max} \le 1$. By Lemma A.17, D has a reachability skeleton R with $|A(R)| \le M(n)$.

We claim that R contains every arc of D. Suppose an arc (u, v) of D is not in R. Since u reaches v in D, it also reaches v in R, along a directed path P. The path P is not the arc (u, v), so it has length at least two. The arc $u \to v$ has no inner vertex, so P and the arc are two internally vertex-disjoint routes. So $\kappa_D(u, v) \ge 2$, a contradiction.

So $|A(D)| = |A(R)| \le M(n)$. With the lower bound, $k_2^{\mathrm{dir}}(n) = M(n)$. □

The arc version follows.

Proof of Theorem 1.7. The two constructions above give $\ell_2^{\mathrm{dir}}(n) \geq M(n)$. Since $\kappa \leq \lambda$, every arc-feasible digraph is vertex-feasible, so Theorem 1.8 bounds it. So

$$M(n) \;\leq\; \ell_2^{\mathrm{dir}}(n) \;\leq\; k_2^{\mathrm{dir}}(n) \;\leq\; M(n),$$

so all three are equal. □

Lemma A.18 (The two directed feasible families at $m = 2$)**.** *A digraph satisfies $\lambda^{\max} \leq 1$ if and only if it satisfies $\kappa^{\max} \leq 1$ under the incidence convention.*

Proof. One direction follows from $\kappa \leq \lambda$. For the other, suppose P_1 and P_2 are two arc-disjoint directed u–v paths. Let w be the first vertex after u on P_1 that also lies on P_2. The parts of P_1 and P_2 from u to w are arc-disjoint and have no inner vertex in common. So the pair (u, w) has two internally disjoint routes. □

Remark A.19 (The directed vertex problem at $m \geq 3$)*.* The skeleton gives the value at $m = 2$ (Theorem 1.8). For $m \geq 3$ the vertex problem is open. The arc constructions give lower bounds on $k_m^{\mathrm{dir}}(n)$.

Lemma A.20 (The augmented bipartite digraph is feasible)**.** *For $m \geq 2$ and $n \geq m$, the digraph of Construction A.15 satisfies $\lambda^{\max} = m - 1$ and has $\lfloor (n+m-2)^2/4 \rfloor$ arcs. Hence $\ell_m^{\mathrm{dir}}(n) \geq \lfloor (n+m-2)^2/4 \rfloor$.*

Proof. Each vertex of B receives one arc from every vertex of A and $m - 2$ arcs from inside B. So the number of arcs is $|B|\,(|A| + m - 2)$. Here $|A| + m - 2 = \lfloor (n+m-2)/2 \rfloor$ and $|B| = \lceil (n+m-2)/2 \rceil$, and $\lfloor N/2 \rfloor \lceil N/2 \rceil = \lfloor N^2/4 \rfloor$ for every integer N.

Now take $a \in A$ and $b \in B$. The direct arc and the $m - 2$ routes $a \to b' \to b$ through the in-neighbours b' of b in B are arc-disjoint. No route from a can enter A again. So every a–b route ends with $a \to b$ or with one of the $m - 2$ arcs into b from B. Removing these $m - 1$ arcs separates a from b, so $\lambda(a, b) = m - 1$. For $b_i, b_j \in B$, a route from b_i stays in B, so it ends with one of the $m - 2$ arcs into b_j from B, and $\lambda(b_i, b_j) \leq m - 2$. There are no routes from B to A or inside A. So $\lambda^{\max} = m - 1$. □

A.6 Structural propositions on the directed frontier

A.6.1 The two-step budget

Definition A.21 (Two-step budget)**.** Let D be a simple digraph and $C \geq 1$. For $u \neq v$, let $p_2(u, v)$ be the number of vertices $x \notin \{u, v\}$ with arcs $u \to x$ and $x \to v$. Such an x is the *midpoint* of the route $u \to x \to v$. We call D *C-budgeted* if every ordered pair (u, v) of distinct vertices satisfies

$$\mathbf{1}_{(u,v) \in A(D)} + p_2(u, v) \;\leq\; C.$$

Lemma A.22 (Feasible digraphs are budgeted)**.** *Let D be a simple digraph and $m \geq 2$. If $\lambda_D^{\max} \leq m-1$, or if $\kappa_D^{\max} \leq m-1$, then D is $(m-1)$-budgeted.*

Proof. Fix an ordered pair (u, v). Take the arc $u \to v$, if it exists, and the route $u \to x \to v$ for each of the $p_2(u, v)$ midpoints x. Two routes with different midpoints use different arcs, and none of them uses the arc $u \to v$. So these routes are arc-disjoint. Each two-step route has one inner vertex, its midpoint, and the arc has none. So the routes are also internally vertex-disjoint. So their number is at most $\lambda_D(u, v)$ and at most $\kappa_D(u, v)$, and either condition gives at most $m-1$. □

The next lemma turns the budget into a statement about degrees. A vertex with large in-degree and large out-degree is the midpoint of many two-step routes, and the budget limits these.

Lemma A.23 (The budget, read on degrees)**.** *Every C-budgeted simple digraph D on n vertices satisfies*

$$\sum_{x \in V} d^+(x)\, d^-(x) \;\leq\; C\, n(n-1).$$

Proof. Summing the budget over all $n(n-1)$ ordered pairs gives $|A(D)| + \sum_{u \neq v} p_2(u, v) \leq C\, n(n-1)$. A *digon* is a pair of vertices with an arc in each direction. Let Δ be the number of digons. We use three facts.

(a) Walks of length two, counted by their midpoint.

$$\sum_{u,\, v \in V} p_2(u, v) \;=\; \sum_{x \in V} d^-(x)\, d^+(x),$$

where the left sum also includes the pairs with $u = v$. Indeed, a walk $u \to x \to v$ is a choice of an in-neighbour u of x and an out-neighbour v of x.

(b) The pairs with $u = v$. The term $p_2(u, u)$ counts walks $u \to x \to u$, and each digon gives two of them. So $\sum_u p_2(u, u) = 2\Delta$.

(c) Different digons use different arcs, so $2\Delta \leq |A(D)|$.

Together with the budget this gives

$$\begin{aligned}
\sum_{x \in V} d^+(x)\, d^-(x) \;&=\; \sum_{u \neq v} p_2(u, v) \;+\; 2\Delta && \text{by (a) and (b),} \\
&\leq\; \big[C\, n(n-1) - |A(D)|\big] \;+\; 2\Delta \\
&\leq\; \big[C\, n(n-1) - |A(D)|\big] \;+\; |A(D)| && \text{by (c),} \\
&=\; C\, n(n-1).
\end{aligned}$$

□

We use two small facts. For $a, b \geq 0$ with $a + b > 0$ we have $\min(a, b) \leq 2ab/(a+b)$, since

$\max(a,b) \geq (a+b)/2$. And $\lfloor (n-1)^2/4 \rfloor + \lfloor n/2 \rfloor = \lfloor n^2/4 \rfloor$, which one checks for even and odd n.

Lemma A.24 (Deleting the smaller half of every vertex)**.** *Every simple digraph D on n vertices can be made* one-way bipartite*, so that all remaining arcs go from one part to the other, by deleting at most $\sum_x \min(d^+(x), d^-(x))$ arcs. So*

$$|A(D)| \;\leq\; \left\lfloor \frac{n^2}{4} \right\rfloor \;+\; \sum_{x \in V} \min\bigl(d^+(x), d^-(x)\bigr).$$

Proof. Call x *source-like* if $d^+(x) \geq d^-(x)$, and *sink-like* otherwise. Let S and T be these two sets. Delete every arc into a source-like vertex and every arc out of a sink-like vertex. At each vertex x this deletes $\min(d^+(x), d^-(x))$ arcs, so in total at most the sum above. An arc $a \to b$ survives only if $a \in S$ and $b \in T$. So the remaining arcs all go from S to T, and there are at most $|S|\,|T| \leq \lfloor n^2/4 \rfloor$ of them. □

Lemma A.25 (The counting core)**.** *Every C-budgeted simple digraph D on $n \geq 2$ vertices satisfies*

$$|A(D)| \;\leq\; \left\lfloor \frac{n^2}{4} \right\rfloor \;+\; 4C(n-1).$$

Proof. We use induction on n. For $n = 2$ a simple digraph has at most 2 arcs, and the right side is $1 + 4C \geq 5$. Let $n \geq 3$.

First suppose some vertex v has total degree $d(v) < n/2$, so $d(v) \leq \lfloor n/2 \rfloor$. Deleting v keeps D budgeted. By induction and the identity above,

$$\begin{aligned} |A(D)| &\leq \left\lfloor \frac{(n-1)^2}{4} \right\rfloor + 4C(n-2) + \left\lfloor \frac{n}{2} \right\rfloor \\ &= \left\lfloor \frac{n^2}{4} \right\rfloor + 4C(n-2) \leq \left\lfloor \frac{n^2}{4} \right\rfloor + 4C(n-1). \end{aligned}$$

Otherwise $d(x) \geq n/2$ for every vertex. By the first fact above and Lemma A.23,

$$\sum_x \min\bigl(d^+(x), d^-(x)\bigr) \;\leq\; \frac{2}{\min_x d(x)} \sum_x d^+(x) d^-(x) \;\leq\; \frac{4}{n}\, C\, n(n-1) \;=\; 4C(n-1),$$

and Lemma A.24 gives the claim. □

We can now prove Theorem 1.12.

Proof of Theorem 1.12. By Lemma A.22 a simple digraph with $\lambda_D^{\max} \leq m-1$ or with $\kappa_D^{\max} \leq m-1$ is $(m-1)$-budgeted, so Lemma A.25 at $C = m-1$ gives the stated bound.

For the lower bound, Construction A.15 has $\lambda^{\max} \leq m-1$, so also $\kappa^{\max} \leq m-1$. For $n \geq m$ this gives

$$k_m^{\mathrm{dir}}(n) \;\geq\; \ell_m^{\mathrm{dir}}(n) \;\geq\; \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor \;=\; \frac{n^2}{4} + \frac{(m-2)n}{2} + O_m(1).$$

So for $m \geq 3$ the second term is linear in n on both sides. □

The linear coefficient is not known. The construction gives $(m-2)n/2$, and Theorem 1.12 allows $4(m-1)n$. For large n, the theorem of Huang and Lyu [12] lowers the upper value to $(m-1)n$.

The next proposition is about the shape of a feasible digraph. After deleting $O(\sqrt{m}\, n^{3/2})$ arcs, all remaining arcs go from one part to the other. The parts need not be balanced. For fixed m this error is larger than the linear error of Theorem 1.12.

Proposition A.26 (Quadratic bound and bipartisation)**.** *Let D be a simple digraph on $n \geq 2$ vertices with $\lambda_D^{\max} \leq m-1$. Then deleting at most $\sqrt{m}\, n^{3/2}$ arcs leaves D a subgraph of a one-directional bipartite digraph, and*

$$|A(D)| \;\leq\; \left\lfloor \frac{n^2}{4} \right\rfloor \;+\; \sqrt{m}\, n^{3/2}.$$

Proof. By Lemma A.22, D is $(m-1)$-budgeted. We use $\min(a,b) \leq \sqrt{ab}$, the Cauchy–Schwarz inequality $\sum_x \sqrt{a_x} \leq \sqrt{n \sum_x a_x}$, and Lemma A.23:

$$\begin{aligned} \sum_{x \in V} \min\bigl(d^+(x), d^-(x)\bigr) \;&\leq\; \sum_{x \in V} \sqrt{d^+(x)\, d^-(x)} \;\leq\; \sqrt{n \sum_{x \in V} d^+(x)\, d^-(x)} \\ &\leq\; \sqrt{n \cdot (m-1)\, n(n-1)} \;=\; n\sqrt{(m-1)(n-1)} \;\leq\; \sqrt{m}\, n^{3/2}. \end{aligned}$$

Lemma A.24 now gives both statements. □

So for $m \geq 3$ both directed problems have leading term $n^2/4$ and a linear second term. The linear coefficient and the exact values are open. It is also open whether the two values are equal.

A.7 The directed multigraph arc formula

The lower bound uses thickened trees and thickened one-way bipartite digraphs. The upper bound uses the reachability skeleton: removing one copy of each skeleton arc lowers every positive connectivity by at least one.

A.7.1 The lower bound: thickening

If we multiply every multiplicity by the same integer, every cut and so every connectivity is multiplied by that integer.

Lemma A.27 (Thickening)**.** *Let D_0 be a directed multigraph and $k \geq 1$ an integer. Write $k \cdot D_0$ for the multigraph obtained by multiplying every multiplicity $\mu(u,v)$ by k. Then $\lambda_{k \cdot D_0}(u,v) = k \cdot \lambda_{D_0}(u,v)$ for every ordered pair $u \neq v$, and $|A(k \cdot D_0)| = k\,|A(D_0)|$.*

Proof. By Menger's theorem, $\lambda_{D_0}(u,v)$ is the smallest number of arcs, counted with multiplicity, in a u–v cut. Multiplying every multiplicity by k multiplies the size of every cut by k. So it multiplies the smallest cut by k too. The number of arcs is also multiplied by k. □

Construction A.28 (Thickened tree and thickened bipartite digraph)**.** Let $m \geq 2$. (a) Take any tree on the n vertices and put $m-1$ arcs in each direction on every tree edge. With one arc in each direction this is the bidirected tree, with $\lambda^{\max} = 1$. So by Lemma A.27 the thickened tree has $\lambda^{\max} = m-1$ and $2(m-1)(n-1)$ arcs. (b) Take the one-way bipartite digraph of Construction A.14 and repeat every arc $m-1$ times. By the same lemma it has $\lambda^{\max} = m-1$ and $(m-1)\lfloor n^2/4 \rfloor$ arcs.

The larger of the two gives $L_m^{\mathrm{dir}}(n) \geq (m-1)M(n)$. We now prove the upper bound.

A.7.2 The upper bound: a reachability skeleton

For the upper bound we want to delete a part of D that lowers every positive connectivity, and whose size we can bound. Deleting one vertex does not work: it need not lower every connectivity. Instead we delete a reachability skeleton. By Lemma A.17 it has at most $M(n)$ arcs, and the next lemma shows that deleting it lowers every positive connectivity.

Lemma A.29 (Deleting a skeleton lowers every positive connectivity)**.** *Let D be a directed multigraph with $\lambda^{\max}(D) \leq k$ for some $k \geq 1$, and let R be a reachability skeleton of D. Let $D-R$ be D with one copy of each arc of R removed. Then for every ordered pair $u \neq v$,*

$$\lambda_{D-R}(u,v) \;\leq\; \max\bigl(\lambda_D(u,v)-1,\,0\bigr),$$

and consequently $\lambda^{\max}(D-R) \leq k-1$.

The skeleton keeps reachability. So if routes from u to v remain after the deletion, one more route runs inside the deleted arcs.

Proof. Fix $u \neq v$ and let $t = \lambda_{D-R}(u,v)$. If $t = 0$ there is nothing to prove. Otherwise $D-R$ has $t \geq 1$ arc-disjoint u–v paths, so u reaches v in D. Since R is a reachability skeleton, there is a u–v path P in R. The arcs of P are the removed copies, so P shares no arc with the t paths in $D-R$. So D has $t+1$ arc-disjoint u–v paths, and $t+1 \leq \lambda_D(u,v)$. Taking the maximum over all pairs gives $\lambda^{\max}(D-R) \leq k-1$. □

Proof of Theorem 1.13. The lower bound is Construction A.28. For the upper bound, fix n. We show by induction on $k \geq 0$ that every directed multigraph D on n vertices with $\lambda^{\max}(D) \leq k$ has $|A(D)| \leq k\,M(n)$. For $k = 0$ there are no arcs, since one arc is already one route.

Let $k \geq 1$, and let R be a reachability skeleton of D with $|A(R)| \leq M(n)$ (Lemma A.17). By Lemma A.29, $\lambda^{\max}(D-R) \leq k-1$, so by induction $|A(D-R)| \leq (k-1)M(n)$. Every arc of D is either in R or in $D-R$. So

$$|A(D)| \;=\; |A(R)| \;+\; |A(D-R)| \;\leq\; M(n) \;+\; (k-1)M(n) \;=\; k\,M(n).$$

With $k = m-1$ this gives $L_m^{\mathrm{dir}}(n) \leq (m-1)M(n)$. □

At $m = 2$ a feasible directed multigraph has no parallel arcs. So Theorem 1.13 at $m = 2$ is Theorem 1.7.

A.7.3 Extremal classification on the quadratic branch

Theorem 1.13 gives the value, but not the extremal multigraphs. This subsection describes them for large n. The idea is to look at the proof of Lemma A.17 in the case of equality. Equality fixes the number of strongly connected components, then the shape of the skeleton, and then the multigraph.

Lemma A.30 (What equality forces)**.** *Let $m \geq 2$ and $n \geq 8$, so that $\lfloor n^2/4 \rfloor > 2(n-1)$ and $M(n) = \lfloor n^2/4 \rfloor$, and let D be a directed multigraph on n vertices with $\lambda_D^{\max} \leq m-1$ and $|A(D)| = (m-1)\lfloor n^2/4 \rfloor$. Then*

(1) D is acyclic, and

(2) D has a reachability skeleton R, inclusion-minimal for reachability in the sense that no arc of it is redundant, whose underlying undirected graph is exactly the balanced complete bipartite graph $K_{\lceil n/2 \rceil, \lfloor n/2 \rfloor}$, so R is an acyclic orientation of it.

Proof. The proof of Theorem 1.13 removes skeletons $R_1, \ldots, R_{m-1}$, and then no arc is left. So $|A(D)| = \sum_i |A(R_i)|$. Each $|A(R_i)| \leq M(n)$, so equality forces $|A(R_i)| = M(n)$ for every i. Let $R := R_1$.

In the proof of Lemma A.17, $|A(R)| \leq f(q) = 2(n-q) + \lfloor q^2/4 \rfloor$, where q is the number of strongly connected components. So $f(q) \geq \lfloor n^2/4 \rfloor = f(n)$. Since $n \geq 8$, $f(1) = 2(n-1) < \lfloor n^2/4 \rfloor = f(n)$. The function f is convex, so for $1 \leq q < n$,

$$f(q) \;\leq\; \frac{(n-q)f(1) + (q-1)f(n)}{n-1} \;<\; f(n).$$

So $q = n$: every strongly connected component is a single vertex. This is (1).

With $q = n$ the trees of Step 1 are empty, so R consists of the arcs chosen for Q, and $|A(Q)| = \lfloor n^2/4 \rfloor$. The graph Q is triangle-free, and a triangle-free graph on n vertices with $\lfloor n^2/4 \rfloor$ edges is $K_{\lceil n/2 \rceil, \lfloor n/2 \rfloor}$ (the equality case of Mantel's theorem [23]). This is (2). The skeleton R is acyclic because D is, and no arc of R can be removed without changing reachability, because the same holds for Q. □

The next lemma uses that no arc of R can be removed.

Lemma A.31 (A minimal bipartite skeleton has no long path)**.** *Let R be as in Lemma A.30, with parts A and B. Then R contains no directed path with three arcs.*

Proof. Suppose $u \to v \to w \to x$ is such a path. The path alternates between the parts, so u and x lie in different parts. The underlying graph of R is complete bipartite, so R has an arc between u and x. It cannot be $x \to u$, since that would close a directed cycle, and D is acyclic.

So $u \to x$ is in R. But u also reaches x along the path, so $u \to x$ could be removed without changing reachability. This is a contradiction. □

Theorem A.32 (Extremal classification on the quadratic branch)**.** *Let $m \geq 2$ and $n \geq \max(8, 2m)$, so that $M(n) = \lfloor n^2/4 \rfloor$ and n is on the quadratic branch. If D is a directed multigraph on n vertices with $\lambda_D^{\max} \leq m-1$ and $|A(D)| = L_m^{\mathrm{dir}}(n) = (m-1)\lfloor n^2/4 \rfloor$, then*

$$D \in \left\{(m-1) \cdot B_{\lceil n/2 \rceil, \lfloor n/2 \rfloor},\ (m-1) \cdot B_{\lfloor n/2 \rfloor, \lceil n/2 \rceil}\right\},$$

where $B_{a,b}$ has all arcs from its part of size a to its part of size b, each once, and $(m-1)\cdot B_{a,b}$ takes every arc $m-1$ times as in Lemma A.27. These two digraphs are the same up to isomorphism when n is even, and reverses of each other when n is odd.

Proof. Take R, A and B from Lemma A.30, and write $\alpha = |A| \geq \beta = |B| = \lfloor n/2 \rfloor$.

Step 1: R has no directed path with two arcs. Suppose $a \to b \to a'$ with $a, a' \in A$ and $b \in B$. If a had an incoming arc $b_0 \to a$, then $b_0 \to a \to b \to a'$ would be a path with three arcs, which Lemma A.31 forbids. So a has no incoming arc. Since the underlying graph is complete bipartite, $a \to b''$ for every $b'' \in B$. In the same way a' has no outgoing arc, so $b'' \to a'$ for every $b'' \in B$. The β routes $a \to b'' \to a'$ use different arcs. So $\lambda_D(a, a') \geq \beta = \lfloor n/2 \rfloor \geq m$, which breaks the condition. A path $b \to a \to b'$ is excluded in the same way, since $\alpha \geq \beta \geq m$. This is the only step that uses $n \geq 2m$.

Step 2: all arcs of R go the same way. By Step 1, no vertex of R has both an incoming and an outgoing arc. Take $a \in A$. It is joined to every vertex of B. If all arcs at a go out, then every $b \in B$ has an incoming arc, so it has no outgoing arc. So all arcs go from A to B. If all arcs at a come in, all arcs go from B to A in the same way. From now on, call the part that the arcs leave A, and the other part B.

Step 3: D has no arc outside R. In R, the only pairs (u, x) with u reaching x are the pairs in $A \times B$, and each such pair is an arc of R. If D has an arc $u \to x$, then u reaches x in D, so also in R. So $u \to x$ is an arc of R.

Step 4: every multiplicity is $m-1$. Parallel arcs are disjoint routes, so every multiplicity is at most $m-1$. There are $|A||B| \leq \lfloor n^2/4 \rfloor$ pairs, and $|A(D)| = (m-1)\lfloor n^2/4 \rfloor$. So every pair has multiplicity $m-1$, and $|A||B| = \lfloor n^2/4 \rfloor$, so the parts have sizes $\lceil n/2 \rceil$ and $\lfloor n/2 \rfloor$. □

The hypothesis $n \geq 2m$ is used only in Step 1. Every vertex of an extremiser has degree at least $(m-1)\lfloor n/2 \rfloor$, because deleting it leaves at most $(m-1)M(n-1)$ arcs. For even n, every vertex has degree exactly $(m-1)n/2$.

Part III. Hypergraphs

The edge bound, the vertex problem at $m = 2$ and $m = 3$, multigraphs with vertex separation, and directed hypergraphs.

A.8 The hypergraph bounds

There are several notions of connectivity in hypergraphs, for example in Chekuri and Xu [25] and Dewar, Pike and Proos [26]. This thesis uses Berge paths throughout. We first need Menger's theorem for Berge paths. It also shows that the gate network of Chapter 2 computes the right value.

Theorem A.33 (Menger for multihypergraphs)**.** *Let u, v be distinct vertices of a multihypergraph $\mathcal{H}$, and count parallel copies of a hyperedge separately. The largest number of u–v Berge paths that share no hyperedge copy equals the smallest number of hyperedge copies whose removal separates u from v.*

Proof. Build the gate network of Figure 2.4b. Each hyperedge e gets an arc $e^- \to e^+$ with capacity $\mu(e)$, and each vertex $x \in e$ gets arcs $x \to e^-$ and $e^+ \to x$ with unlimited capacity. We show that disjoint Berge paths correspond to flows, and that cuts correspond to cuts.

From paths to flow. A Berge path $u, e_1, v_1, \dots, e_k, v$ becomes the walk $u \to e_1^- \to e_1^+ \to v_1 \to \cdots \to v$, carrying one unit of flow. In a family of paths that share no copy, at most $\mu(e)$ paths use the hyperedge e. So the family gives a flow whose value is the number of paths.

From flow to paths. All capacities are integers, so there is a maximum flow with integer values, and it splits into unit paths. Replace each step $x \to e^- \to e^+ \to y$ by the hyperedge step x, e, y. This turns each unit path into a Berge path. At most $\mu(e)$ of them use e, so each can get its own copy of e.

Cuts. A cut of finite capacity uses only the arcs $e^- \to e^+$. Such a set of arcs separates u from v exactly when the corresponding hyperedges separate them in $\mathcal{H}$, and its capacity is the number of copies. Removing only some copies of a hyperedge changes nothing, so a smallest cut in $\mathcal{H}$ removes all copies of some hyperedges. So the two notions of cut have the same minimum, and the max-flow min-cut theorem gives the claim. □

Lemma A.34 (Cut induction for multihypergraphs)**.** *Let $\mathcal{H}$ be a multihypergraph on $n \geq 1$ vertices, not assumed uniform, with $\lambda_{\mathcal{H}}^{\max} \leq k$. Then, the sum running over hyperedge copies,*

$$\sum_{e} \big(|e| - 1\big) \;\leq\; k\,(n-1).$$

Proof. Induction on n. At $n = 1$ every hyperedge is a single vertex and both sides are 0.

Let $n \geq 2$ and fix two vertices $u \neq v$. By Theorem A.33, a set F of at most k hyperedge

copies separates them. Let S be the set of vertices that u can reach in $\mathcal{H} - F$, and let $\bar{S} = V \setminus S$. Then $u \in S$ and $v \in \bar{S}$, and every copy that meets both sides lies in F. So at most k copies meet both sides.

Form two *trace* multihypergraphs $\mathcal{H}[S]$ and $\mathcal{H}[\bar{S}]$: replace every copy e by $e \cap S$, or by $e \cap \bar{S}$, and drop the sets with fewer than two vertices. A Berge route in a trace is also a Berge route in $\mathcal{H}$, using the original copies. So a family of disjoint routes in a trace gives one of the same size in $\mathcal{H}$. So both traces have $\lambda^{\max} \le k$, and the induction hypothesis applies to both.

Now count. A copy inside one side contributes $|e| - 1$ to that side. A copy that meets both sides satisfies

$$|e| - 1 \;=\; \big(|e \cap S| - 1\big) \;+\; \big(|e \cap \bar{S}| - 1\big) \;+\; 1,$$

where each bracket is the contribution to one trace (zero if that part was dropped). Summing over all copies gives

$$\sum_e \big(|e| - 1\big) \;\le\; k\big(|S| - 1\big) + k\big(|\bar{S}| - 1\big) + \big|\{\text{crossing copies}\}\big| \;\le\; k(n-2) + k \;=\; k(n-1). \qquad \square$$

A *hypertree* is a hypergraph whose incidence graph (Section 1.3.1) is a tree.

Proof of Theorem 1.9. Upper bound. Every hyperedge has $|e| = r$, so Lemma A.34 with $k = m - 1$ gives $(r-1)|E(\mathcal{H})| \le (m-1)(n-1)$. Since $|E(\mathcal{H})|$ is an integer, we can round down.

Lower bound. Suppose $r - 1$ divides $n - 1$. Take a hub h and split the other $n - 1$ vertices into groups $G_1, G_2, \ldots$ of size $r - 1$. The hyperedges are the sets $\{h\} \cup G_i$. Its incidence graph is a tree, so we call it a *star hypertree*. Give each hyperedge multiplicity $m - 1$. Every vertex other than h lies in exactly $m-1$ hyperedge copies, and removing them isolates it. So by Theorem A.33 every pair has $\lambda \le m - 1$. There are $(m-1)(n-1)/(r-1)$ hyperedges.

If $m - 1 \le \binom{n-2}{r-2}$ and $r \ge 3$, Theorem A.35 below attains the bound with a simple hypergraph. If $r = 2$, this condition means $m = 2$. Then the bound is $n - 1$, and a tree attains it. $\square$

The next theorem shows that for $r \ge 3$ and $m - 1 \le \binom{n-2}{r-2}$, simple hypergraphs attain the same bound. For graphs ($r = 2$) this is different: there, simplicity lowers the bound from $(m-1)(n-1)$ to $\lfloor m(n-1)/2 \rfloor$.

Theorem A.35 (Simplicity is free for $r \ge 3$)**.** *Let $r \ge 3$, $n \ge r$, and $2 \le m$ with $m - 1 \le \binom{n-2}{r-2}$. Then the maximum number of hyperedges in a* simple *r-uniform hypergraph on n vertices with $\lambda^{\max} \le m - 1$ is exactly $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$.*

Proof. The upper bound is Theorem 1.9. For the lower bound, take a hub h and let $V' = V \setminus \{h\}$. By Lemma A.38 with uniformity $r - 1$ and degree bound $d = m - 1$ on the $n - 1$ vertices of V', there is a simple $(r-1)$-uniform hypergraph $\mathcal{G}^*$ on V' with maximum degree at most $m - 1$ and $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ hyperedges. Add h to each of its hyperedges. The new hyperedges are different r-sets, so the result $\mathcal{H}$ is simple. Every vertex $w \ne h$ lies in at most $m - 1$ hyperedges, and removing them isolates w. So by Theorem A.33, $\lambda_{\mathcal{H}}^{\max} \le m - 1$. $\square$

Remark A.36 (The condition is needed). Without the condition $m - 1 \leq \binom{n-2}{r-2}$, simple hypergraphs may not reach the bound. The smallest example is $n = r = m = 3$. There is only one 3-set, so a simple hypergraph has at most one hyperedge, while the bound is $\lfloor 2 \cdot 2/2 \rfloor = 2$. Two copies of the one hyperedge do reach the bound. Theorem 1.11 needs the same condition with $m - 1 = 2$, which is the range $2 < r < n$.

Construction A.37 (Twelve hyperedge copies on six vertices)**.** Take a hub h and five other vertices a, b, c, d, e. Include all ten triples consisting of h and two of the other vertices. Add a second copy of $\{h, d, e\}$ and one copy of $\{a, b, c\}$. The resulting 3-uniform multihypergraph has 12 hyperedge copies and $\lambda^{\max} \leq 5$. Consequently, the multihypergraph edge value at $(n, m, r) = (6, 6, 3)$ is exactly 12, even though $(r - 1)$ does not divide $(n - 1)$.

Proof. Initially each non-hub vertex lies in four of the ten triples. The repeated triple raises the degrees of d and e to five, while the final triple does the same for a, b and c. Every pair of distinct vertices therefore includes a non-hub vertex whose five incident hyperedge copies form a cut separating it from the other endpoint. Thus no pair has more than five hyperedge-disjoint routes. The construction has $10+1+1 = 12$ copies, matching the upper bound $\lfloor 5(6-1)/(3-1) \rfloor = 12$ from Theorem 1.9. □

Since $\kappa \leq \lambda$, the same multihypergraph also gives 12 for the vertex problem. We do not know whether 12 is optimal there.

The proof of Theorem A.35 uses the following lemma.

Lemma A.38 (Sparse uniform hypergraphs exist)**.** *Let r, n and d be integers with $r \geq 1$, $n \geq r$ and $0 \leq d \leq \binom{n-1}{r-1}$. There is a simple r-uniform hypergraph on n vertices with exactly $\lfloor \frac{dn}{r} \rfloor$ hyperedges and maximum degree at most d.*

Proof. Let $N := \lfloor \frac{dn}{r} \rfloor$. Then $N \leq \frac{n}{r}\binom{n-1}{r-1} = \binom{n}{r}$, so there are at least N different r-sets. Among all families of N different r-sets, choose one with the smallest value of $\sum_x \deg(x)^2$. Suppose $\deg(x) \geq \deg(y) + 2$ for two vertices x, y. Let a be the number of sets in the family that contain x but not y, and b the number that contain y but not x. Then $a \geq b + 2$. The map $A \mapsto A - x + y$ sends the first kind of set to the second kind, and it is injective. Since $a > b$, some image $A - x + y$ is not in the family. Replace A by $A - x + y$. The family stays simple and has N sets, and the sum changes by

$$(\deg(x) - 1)^2 + (\deg(y) + 1)^2 - \deg(x)^2 - \deg(y)^2 = 2\big(\deg(y) - \deg(x)\big) + 2 \leq -2.$$

This contradicts the choice of the family. So any two degrees differ by at most one. The degrees add up to $rN \leq dn$, so the largest degree is $\lceil rN/n \rceil \leq d$. □

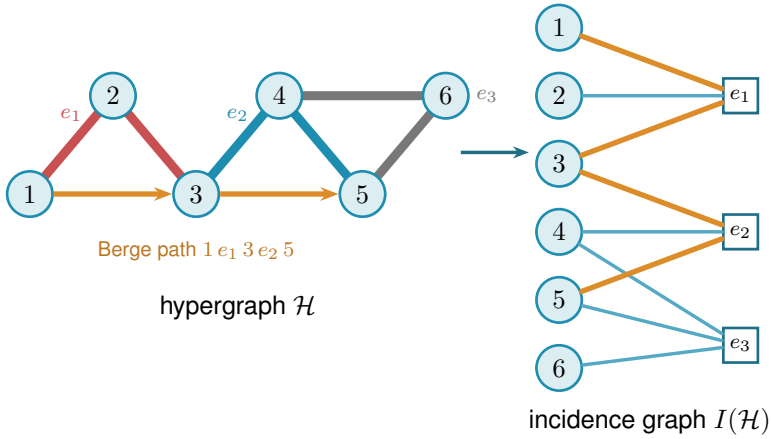
A.9 The hypergraph vertex problem

The vertex problem for hypergraphs becomes a problem about graphs (Figure A.3). Recall the *incidence graph* $I(\mathcal{H})$: one node per vertex, one node per hyperedge, and an edge when the

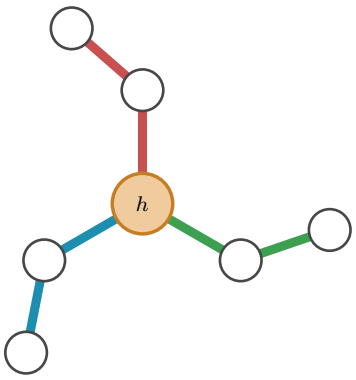
vertex lies in the hyperedge. A Berge path is the same as a path in $I(\mathcal{H})$ between two vertex nodes. Its inner nodes are the hyperedges it uses and the vertices where it changes hyperedge. So two Berge routes share no hyperedge and no inner vertex exactly when the two paths in $I(\mathcal{H})$ are internally disjoint. So

$$\kappa_{\mathcal{H}}(u, v) \;=\; \kappa_{I(\mathcal{H})}(u, v).$$

The vertex problem asks for the largest number of hyperedge nodes, each of degree r, such that no two of the n vertex nodes are joined by m internally disjoint paths.



(a) A 3-uniform hypergraph $\mathcal{H}$ (left) and its bipartite incidence graph $I(\mathcal{H})$ (right): circles for vertices, hollow squares for hyperedges, an edge when the vertex lies in the hyperedge. The Berge path $1\, e_1\, 3\, e_2\, 5$ is traced in orange in both pictures. In $\mathcal{H}$ it rides the red line e_1 and the blue line e_2, changing at the shared vertex 3, while the grey line e_3 lies off it. In $I(\mathcal{H})$ it is the ordinary path $1-e_1-3-e_2-5$ alternating sides.



(b) A star hypertree: every hyperedge $\{h\} \cup G_i$ shares only the hub h (orange), and the other $r - 1 = 2$ vertices of each block sit in a single hyperedge. Its incidence graph is a tree, so $\kappa^{\max} \le 1$.

Figure A.3. The translation the vertex problem rests on. Internally vertex-disjoint Berge paths in $\mathcal{H}$ correspond exactly to internally disjoint paths in $I(\mathcal{H})$, so $\kappa_{\mathcal{H}}(u, v) = \kappa_{I(\mathcal{H})}(u, v)$. **(top)** A hypergraph and its incidence graph with one Berge path drawn in both. **(bottom)** The star hypertree of Theorem 1.10, whose incidence graph is a tree.

We now prove Theorem 1.10.

Proof of Theorem 1.10. We claim that $\kappa_{\mathcal{H}}^{\max} \le 1$ exactly when $I(\mathcal{H})$ is a forest. Suppose $I(\mathcal{H})$ has a cycle $v_1, e_1, v_2, e_2, \ldots, v_\ell, e_\ell, v_1$, with $\ell \ge 2$ and all v_i and e_i distinct. Then v_1, e_1, v_2 and $v_1, e_\ell, v_\ell, \ldots, e_2, v_2$ are two Berge routes from v_1 to v_2 that share no hyperedge and no inner vertex. So $\kappa(v_1, v_2) \ge 2$. Conversely, two such routes between u and v together form a cycle in

$I(\mathcal{H})$. This proves the claim.

A forest on $n+q$ nodes has at most $n+q-1$ edges, and $I(\mathcal{H})$ has rq edges. So $rq \leq n+q-1$, that is, $q \leq (n-1)/(r-1)$. The star hypertree has $\lfloor (n-1)/(r-1) \rfloor$ hyperedges. Its hyperedges meet only at the hub, and every other vertex lies in at most one hyperedge, so its incidence graph is a forest. □

Proposition A.39 (General lower bound)**.** *For $r \geq 3$, $m \geq 2$ and $m-1 \leq \binom{n-2}{r-2}$,*

$$k_m^{(r)}(n) \;\geq\; \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor,$$

attained by a simple hypergraph.

Proof. The simple hypergraph of Theorem A.35 has $\lambda^{\max} \leq m-1$. Since $\kappa \leq \lambda$, it is also feasible for the vertex problem. □

That construction needs different $(r-1)$-sets, which explains the condition $m-1 \leq \binom{n-2}{r-2}$. A multihypergraph can repeat a hyperedge instead. This covers other cases.

Proposition A.40 (The multihypergraph lower bound)**.** *For $r \geq 2$, $m \geq 2$ and $(r-1) \mid (n-1)$,*

$$K_m^{(r)}(n) \;\geq\; \frac{(m-1)(n-1)}{r-1},$$

attained by the star hypertree of Figure 1.8 with every hyperedge taken at multiplicity $m-1$. At $m=3$ this meets the upper bound of Theorem 1.11, so

$$K_3^{(r)}(n) \;=\; \left\lfloor \frac{2(n-1)}{r-1} \right\rfloor \qquad \textit{whenever } (r-1) \mid (n-1).$$

Proof. Take the hub h and split the other $n-1$ vertices into groups $G_1, \ldots, G_q$ of size $r-1$, where $q = (n-1)/(r-1)$. Give each hyperedge $\{h\} \cup G_i$ multiplicity $m-1$. This gives $(m-1)q$ hyperedges.

Now take two vertices $u \neq w$. Suppose first that both lie in $\{h\} \cup G_i$. Then the $m-1$ copies of $\{h\} \cup G_i$ are $m-1$ separated routes of length one, so $\kappa(u,w) \geq m-1$. One of u, w lies in G_i, and these $m-1$ copies are all hyperedges that contain it. So removing them separates u from w, and $\kappa(u,w) \leq \lambda(u,w) \leq m-1$ by Theorem A.33. Now suppose $u \in G_i$ and $w \in G_j$ with $i \neq j$. Every route from u to w passes through h, so $\kappa(u,w) \leq 1$. So the hypergraph is feasible.

At $m=3$ there are $2(n-1)/(r-1)$ hyperedges, which equals the upper bound of Theorem 1.11. □

The upper bound at $m=3$ follows from a bound on the cycles of a graph, proved next. For a connected multigraph G, the *cycle rank* is $\operatorname{rank}(G) = |E(G)| - |V(G)| + 1$. A tree has cycle rank 0, and each extra edge raises it by one.

We use the words k-connected, cut vertex, separating pair and block from Section A.2.1. By Menger's theorem [4, Ch. 3], a graph is k-connected exactly when every two vertices are joined by k internally disjoint paths. In a multigraph, a block is a bridge, a set of parallel edges between two vertices, or a maximal 2-connected subgraph. Two blocks share at most one vertex, which is a cut vertex. The blocks and the cut vertices form a tree, the *block-cut tree*: it has a node for each block and for each cut vertex, and a cut vertex is joined to the blocks that contain it.

Lemma A.41 (Incidence rank lemma)**.** *Let G be a connected loopless multigraph with a partition $V(G) = X \dot\cup Z$ such that (i) Z is independent, (ii) no edge incident to Z is parallel to another edge, (iii) every $z \in Z$ has degree at least 2, and (iv) $\kappa_G(x, x') \le 2$ for all distinct $x, x' \in X$. Then $\operatorname{rank}(G) \le |X| - 1$.*

Read X as the vertices of a hypergraph and Z as its hyperedges, so that G is the incidence graph. Then (i) says that two hyperedge nodes are never adjacent, (ii) that no hyperedge repeats a vertex, (iii) that every hyperedge has at least two vertices, and (iv) that the hypergraph is feasible at $m = 3$. We allow edges inside X, because Step 2 of the proof creates them.

Proof. We use induction on $|V|+|E|$. Step 1 handles cut vertices. Steps 2 and 3 remove vertices of degree two. Step 4 handles what is left: a 2-connected graph with minimum degree at least three.

Base: $|V| \le 2$. A vertex in Z has degree at least 2 by (iii), and by (ii) its edges go to different vertices. So with at most two vertices, Z is empty and all vertices lie in X. With one vertex, $\operatorname{rank} = 0 = |X| - 1$. With two vertices, the parallel edges between them are internally disjoint routes, so by (iv) there are at most two, and $\operatorname{rank} \le 1 = |X| - 1$.

Step 1: a cut vertex. Suppose G has a cut vertex. Let $B_1, \dots, B_c$ be its blocks, with $c \ge 2$. A cycle cannot pass through two blocks: it would leave a block and return through the same cut vertex. So $\operatorname{rank}(G) = \sum_i \operatorname{rank}(B_i)$. It is enough to show $\operatorname{rank}(B_i) \le |X \cap B_i| - 1$ for each block and add up.

A bridge has rank 0, and by (i) at least one of its ends lies in X, so $0 \le |X \cap B| - 1$. Every other block has minimum degree at least 2, so (i) to (iv) hold in it. By induction, $\operatorname{rank}(B_i) \le |X \cap B_i| - 1$.

When we add up, a cut vertex is counted once for each block that contains it. Let $b(v)$ be the number of blocks that contain v. Then

$$\sum_i |X \cap B_i| = |X| + \sum_{x \in X \text{ cut}} (b(x) - 1).$$

The block-cut tree has c block nodes and one node per cut vertex. It is a tree, so it has $c + \#\{\text{cut vertices}\} - 1$ edges. Each cut vertex v is joined to $b(v)$ blocks, so it also has $\sum_{v \text{ cut}} b(v)$ edges. So $c = 1 + \sum_{v \text{ cut}} (b(v) - 1)$. Together,

$$\operatorname{rank}(G) = \sum_i \operatorname{rank}(B_i) \le \sum_i (|X \cap B_i| - 1) = |X| - 1 - \sum_{z \in Z \text{ cut}} (b(z) - 1) \le |X| - 1.$$

Step 2: a Z-vertex of degree two. From now on G is 2-connected. Suppose some $z_0 \in Z$ has degree 2. By (i) its two neighbours x, x' lie in X, and by (ii) they are different. *Suppress* z_0: delete it and add a new edge xx'. Then $|E|$ and $|V|$ both drop by one, so the rank does not change. The new edge lies inside X, so $|X|$ does not change, and (i), (ii), (iii) still hold. A route through z_0 becomes a route through the new edge, so every $\kappa(x_1, x_2)$ stays the same and (iv) holds. The graph is smaller, so the induction hypothesis gives the bound.

Step 3: an X-vertex of degree two. Now every Z-vertex has degree at least 3. Suppose some $x_0 \in X$ has degree 2, and delete it. Since G is 2-connected, the result is connected. We remove two edges and one vertex, so the rank drops by one, and $|X|$ drops by one. A neighbour in Z keeps degree at least 2, so (iii) holds. Deleting a vertex cannot raise a connectivity, so (iv) holds. Also $|X| \geq 2$: otherwise, by (i) and (ii), every z would have degree at most 1. By induction, $\mathrm{rank}(G) - 1 \leq (|X| - 1) - 1$.

Step 4: the rest. Now G is 2-connected, has at least three vertices, and every degree is at least 3.

First, G is simple. By (ii), parallel edges can only join two vertices x, x' of X. Suppose $\mu(x, x') \geq 2$. Take a third vertex w. Since G is 2-connected, $G - x'$ has a walk from x to w, and $G - x$ has a walk from w to x'. Together they form an x–x' walk through w. After its last visit to x it continues inside $G - x'$ or at w, so its next vertex is not x'. So the walk contains an x–x' path with at least one inner vertex. With the two parallel edges this gives $\kappa(x, x') \geq 3$, against (iv). So G is simple, and since every degree is at least three, $|V| \geq 4$.

If G is 3-connected, then every pair has $\kappa \geq 3$. By (iv), X then has at most one vertex, and by (i) every z has degree at most 1, which is impossible.

So G has a separating pair $\{a, b\}$. Remove it, and divide the remaining components into two non-empty groups with vertex sets U and W. Since G is 2-connected, a and b both have neighbours in U and in W, and each side contains an a–b path. Let $s = |X \cap \{a, b\}|$. We build two smaller graphs G_U and G_W, the *torsos*. G_U is the subgraph on $U \cup \{a, b\}$ with one extra a–b connection, and G_W is built in the same way:

- if $ab \in E(G)$, retain that edge on both sides.
- if $ab \notin E(G)$ and $s \geq 1$, add one new edge ab to each side.
- if $s = 0$, add to each side a fresh X-vertex together with the two edges joining it to a and to b.

These cases cover everything: by (i), if $ab \in E(G)$ then $s \geq 1$. In that case $s = 1$. Indeed, if $a, b \in X$, the edge ab and one a–b path on each side would give $\kappa(a, b) \geq 3$.

Each torso satisfies (i) to (iv). (i): every added edge has an end in X. In the third case this is the new vertex. (ii): an added edge joins two vertices that had no edge. (iii): a Z-vertex in U keeps all its neighbours. If a or b lies in Z, it keeps a neighbour on its side and gains the added connection. (iv): a new X-vertex has degree two, so its connectivity is at most two. For two old X-vertices, internally disjoint paths can share no edge except xx' itself, so at most one path

uses the added connection. Replace it by an a–b path through the other side. So three internally disjoint paths in a torso would give three in G, which (iv) forbids.

Both torsos are smaller in $|V| + |E|$. Indeed, neither side is a single vertex: in a simple graph it would have only a and b as neighbours, but every degree is at least three. So a torso loses at least two vertices and three edges, and gains at most one vertex and two edges.

Counting. Let r_U, r_W be the ranks of the torsos and X_U, X_W their X-sets. From $\mathrm{rank} = |E| - |V| + 1$:

- first case: $\mathrm{rank}(G) = r_U + r_W$ and $|X_U| + |X_W| = |X| + 1$.
- second case: $\mathrm{rank}(G) = r_U + r_W - 1$ and $|X_U| + |X_W| = |X| + s$.
- third case: $\mathrm{rank}(G) = r_U + r_W - 1$ and $|X_U| + |X_W| = |X| + 2$.

By induction $r_U \le |X_U| - 1$ and $r_W \le |X_W| - 1$. This gives $\mathrm{rank}(G) \le |X| - 1$ in the first and third cases, and $\mathrm{rank}(G) \le |X| + s - 3 \le |X| - 1$ in the second, since $s \le 2$. □

We now prove Theorem 1.11.

Proof of Theorem 1.11. Apply Lemma A.41 to each component of the incidence graph $I(\mathcal{H})$, with X the vertex nodes and Z the hyperedge nodes. The conditions hold: Z is independent, a hyperedge contains each vertex once, hyperedge nodes have degree $r \ge 2$, every component contains a vertex node, and $\kappa_I(u, v) = \kappa_{\mathcal{H}}(u, v) \le 2$. Let C be the number of components. $I(\mathcal{H})$ has rq edges and $n + q$ nodes, so the ranks add up to $rq - (n + q) + C$. The bounds add up to $n - C$. So $rq - (n + q) + C \le n - C$, that is, $q(r - 1) \le 2(n - 1) - 2(C - 1) \le 2(n - 1)$. The lower bound is Proposition A.39 at $m = 3$. □

Remark A.42 (Edges as gates at $r = 2$, and the boundary of the method). At $r = 2$ the theorem is about multigraphs. It gives $K_3(n) = 2(n - 1)$ (Corollary A.46). The proof of Lemma A.41 uses the condition $\kappa \le 2$ in several places. For example, two parallel edges and one more route already break it, and so does a 3-connected graph. The lemma is false for $\kappa \le 3$: two X-vertices joined by three parallel edges have rank two, while $|X| - 1 = 1$. For $m = 4$ a different bound would be needed, probably with a decomposition along sets of three vertices. We do not have one.

A.10 The multigraph incidence problems

At $r = 2$ the hypergraph vertex problems are about multigraphs. They are not the same as the simple problems. Recall that q parallel copies of uv count as q separated routes. Write $K_m(n)$ for the largest number of edges, counted with multiplicity, of a multigraph on n vertices with $\kappa^{\max} \le m - 1$.

The parallel edges and the longer routes do not interfere, so we can count them separately.

Lemma A.43 (Splitting the vertex measure)**.** *Let G be a multigraph with multiplicity function μ and underlying simple graph G_0. For any two vertices $u \neq v$, write $\pi(u,v)$ for the largest number of pairwise internally disjoint u–v paths of length at least two in G_0. Then*

$$\kappa_G(u,v) \;=\; \mu(u,v) \;+\; \pi(u,v).$$

Proof. A route of length one is a copy of the edge uv. It has no inner vertex, so it is separated from every other route.

Lower bound. Take all $\mu(u,v)$ copies of uv, and $\pi(u,v)$ internally disjoint u–v paths of length at least two in G_0. Use one copy of each edge on these paths. The inner vertices do not change, so all these routes are separated.

Upper bound. Take a family of separated u–v routes in G. At most $\mu(u,v)$ of them have length one, since they use different copies of uv. Map each longer route to its sequence of vertices, a path in G_0. Two longer routes have disjoint, non-empty sets of inner vertices. So their images are different and internally disjoint, and there are at most $\pi(u,v)$ of them. □

So a multigraph is feasible exactly when $\mu(u,v) \leq m-1-\pi(u,v)$ on every edge and $\pi(u,v) \leq m-1$ on every pair.

Theorem A.44 (The multigraph incidence value at $m=2$)**.** $K_2(n) = n-1$*, attained exactly by the spanning trees.*

Proof. By Lemma A.43, $\mu(u,v)+\pi(u,v) \leq 1$ on every edge. If G_0 had a cycle, an edge uv on it would have $\mu(u,v) \geq 1$ and $\pi(u,v) \geq 1$. So G_0 is a forest and every multiplicity is one. A forest has at most $n-1$ edges, and a spanning tree has exactly $n-1$. □

Remark A.45. At $m=2$ a second copy of an edge is already a second route, so every feasible multigraph is simple. So $K_2(n) = k_2(n) = n-1$. This is also Theorem 1.10 at $r=2$.

The value at $m=3$ follows from the hypergraph theorem.

Corollary A.46 (The multigraph incidence value at $m=3$)**.** $K_3(n) = 2(n-1)$ *for every* $n \geq 2$*.*

Proof. Apply Theorem 1.11 and Proposition A.40 at $r=2$. The divisibility condition is automatic, and the doubled star attains the upper bound $2(n-1)$. □

So at $m=3$ the vertex value equals the edge value $L_3(n)$ of Theorem 1.4. At $m=5$ they differ: the thickened theta below gives $K_5(4) \geq 14$, while $L_5(4) = 12$. For $m=4$ this is open. For $m \geq 3$ there are three constructions.

★ *The thickened tree.* Take a spanning tree and give every edge multiplicity $m-1$. Tree edges have $\pi = 0$, and other pairs have $\pi = 1$, so it is feasible. It has $(m-1)(n-1)$ edges, the edge value of Theorem 1.4.

- ★ *The thickened complete graph.* Give every edge of K_n multiplicity q. Here $\pi(u,v) = n-2$, so it is feasible when $q \le m+1-n$. It has $\binom{n}{2}(m+1-n)$ edges, which is positive only when $n \le m$.

- ★ *The thickened theta.* Choose two *poles*. Join each of the other $n-2$ vertices to both poles with multiplicity $m-2$, and join the poles with multiplicity $m+1-n$ if this is positive. A pole and another vertex have $\pi = 1$, two non-poles have $\pi = 2$, and the two poles have $\pi = n-2$. So it is feasible when $n \le m+1$, and it has $2(n-2)(m-2) + \max(0,\, m+1-n)$ edges.

For $n \le m+1$, the theta has as many edges as the thickened tree at $m = 4$, and more for $m \ge 5$ (when $n \ge 3$). For $n \ge m+2$ the other two constructions are not feasible, so among these three the tree is best.

For $m \ge 5$, one triangle added to the thickened tree already beats it, also when $n \ge m+2$. The next result makes this precise.

Fix a connected simple graph G_0. By Lemma A.43, the best multigraph on G_0 takes $\mu(u,v) = m-1-\pi(u,v)$ on every edge. Its size is $(m-1)\,|E(G_0)| - \sum_{uv\in E(G_0)} \pi(u,v)$. Since $|E(G_0)| = n-1+\operatorname{rank}(G_0)$, it beats the thickened tree exactly when

$$\sum_{uv\in E(G_0)} \pi(u,v) \;<\; (m-1)\operatorname{rank}(G_0).$$

Theorem A.47 (A bouquet of thickened cliques, and how it beats the thickened tree)**.** *Let $3 \le s \le m$ and $q = m+1-s \ge 1$. Let $1 \le k \le \lfloor (n-1)/(s-1) \rfloor$. Take k copies of K_s that share one vertex and are otherwise disjoint, and give every edge multiplicity q. Attach the remaining vertices as a path with every edge of multiplicity $m-1$. This multigraph has $\kappa^{\max} \le m-1$, and its number of edges is*

$$(m-1)(n-1) \;+\; k\cdot \operatorname{gain}(s,m), \qquad \operatorname{gain}(s,m) := \frac{(s-1)(s-2)(m-1-s)}{2}.$$

So it beats the thickened tree exactly when $s \le m-2$. Such an s exists when $m \ge 5$.

Proof. Write G_0 for the underlying simple graph.

Feasibility. The copies of K_s and the path edges meet only in cut vertices. A path between two vertices of the same block cannot leave the block, since it would pass a cut vertex twice. Take an edge uv in a copy of K_s. The other $s-2$ vertices of the copy give $s-2$ internally disjoint routes of length two, and they also form a cut, so $\pi(u,v) = s-2$. By Lemma A.43, $\kappa_G(u,v) = q + (s-2) = m-1$. Two vertices in different blocks are separated by a cut vertex, so $\kappa_G(u,v) \le 1$. For an edge uv on the path, $\pi(u,v) = 0$, so $\kappa_G(u,v) = m-1$. Inside a block every two vertices are adjacent, so no other pairs need checking.

Count. Since $q \ge 1$, the underlying simple graph G_0 is connected, so $|E(G_0)| = n - 1 + \operatorname{rank}(G_0)$. Only the k copies of K_s contain cycles or have $\pi > 0$. Each copy has rank

$(s-1)(s-2)/2$, and its $\binom{s}{2}$ edges each have $\pi = s-2$. So the number of edges is

$$(m-1)\Big[n-1+k\tfrac{(s-1)(s-2)}{2}\Big] - k\tfrac{s(s-1)(s-2)}{2} \;=\; (m-1)(n-1) + k\,\mathrm{gain}(s,m).$$

□

At $s=3$, $\mathrm{gain}(3,m) = m-4$. This is zero at $m=4$ and positive for $m \geq 5$. So for every $m \geq 5$ the thickened tree is not optimal, also when $n \geq m+2$. Here is the smallest example. Take $m=5$ and $n=7$. Put a triangle on $\{u_1,u_2,u_3\}$ with multiplicity 3, and attach a path of four more vertices to u_1 with multiplicity 4. Two triangle vertices have three parallel edges and one route through the third vertex, so $\kappa = 4$. Every other pair is separated by a cut vertex or joined only by its parallel edges. So $\kappa^{\max} = 4 = m-1$. There are $3\cdot 3 + 4\cdot 4 = 25$ edges, one more than the thickened tree's 24. With k triangles the gain is k, so the gap grows linearly in n.

Sharing a vertex lets more copies fit, since each new copy uses only $s-1$ new vertices. For example, at $m=10$, $s=6$ and $n=11$, two copies sharing a vertex give 150 edges. Disjoint copies would allow only one copy, with 75 edges, and 120 edges after attaching the other five vertices.

The best s is about $(m+1)/2$, where the gain per added vertex, $\mathrm{gain}(s,m)/(s-1)$, is of order m^2. So for $n/m \to \infty$ the construction gives $K_m(n) \geq \Theta(m^2)\,n$, while the thickened tree gives only $(m-1)(n-1)$.

This construction need not be optimal. The next result gives an upper bound of the same order.

Proposition A.48 (An upper bound of the same order)**.** *For all $m \geq 2$ and $n \geq 2$,*

$$K_m(n) \;\leq\; (m-1)\,k_m(n) \;<\; 2(m-1)^2\,n.$$

Proof. Let G be feasible, with underlying simple graph G_0.

Every multiplicity is at most $m-1$. By Lemma A.43, $\kappa_G(u,v) = \mu(u,v) + \pi(u,v) \geq \mu(u,v)$, so feasibility gives $\mu(u,v) \leq m-1$ on every pair and the total is at most $(m-1)|E(G_0)|$.

G_0 is itself feasible for the simple vertex problem. Take two vertices $u \neq v$. If they are adjacent in G_0 then $\mu(u,v) \geq 1$, and the routes of G_0 between them are the single edge together with the internally disjoint routes of length at least two, so $\kappa_{G_0}(u,v) = 1 + \pi(u,v) \leq \mu(u,v) + \pi(u,v) \leq m-1$. If they are not adjacent then $\kappa_{G_0}(u,v) = \pi(u,v) \leq m-1$ directly. So $\kappa_{G_0}^{\max} \leq m-1$ and $|E(G_0)| \leq k_m(n)$ by definition of k_m, which gives the first inequality.

The simple problem is linearly bounded. Suppose G_0 had average degree at least $4(m-1)$. By Mader's density theorem [21], in the form of [4, Theorem 1.4.3], a graph of average degree at least $4k$ contains a $(k+1)$-connected subgraph, so with $k = m-1$ some subgraph $H \subseteq G_0$ is m-connected. An m-connected graph has more than m vertices and, by the global form of Menger's theorem [4, Ch. 3], any two of them are joined by m internally disjoint paths inside H and hence inside G_0. That contradicts $\kappa_{G_0}^{\max} \leq m-1$. So the average degree is below $4(m-1)$,

that is $2|E(G_0)|/n < 4(m-1)$, giving $k_m(n) < 2(m-1)n$ and the second inequality. □

So for $n/m \to \infty$ and then $m \to \infty$,

$$\left(\tfrac{1}{8} + o_m(1)\right) m^2 n \;\le\; K_m(n) \;<\; 2m^2 n.$$

This needs n large: for example $K_m(2) = m-1$. Theorem A.51 below gives a better lower constant. The first inequality, $K_m(n) \le (m-1)k_m(n)$, says that $K_m(n)$ is at most $m-1$ times the simple vertex value. It is tight at $m = 2$ (Remark A.45).

A.10.1 The value is a block problem

The next two results reduce $K_m(n)$ to 2-connected blocks, as in Section A.2.1.

Lemma A.49 (The objective in closed form)**.** *For a simple graph G_0 on n vertices with $\kappa^{\max}_{G_0} \le m-1$, the largest total multiplicity of a feasible multigraph with underlying simple graph exactly G_0 is*

$$W_m(G_0) \;:=\; \sum_{uv \in E(G_0)} \big(m - \kappa_{G_0}(u,v)\big),$$

and every simple graph arising as the underlying graph of a feasible multigraph satisfies $\kappa^{\max}_{G_0} \le m-1$. Hence

$$K_m(n) \;=\; \max\big\{\, W_m(G_0) \;:\; G_0 \text{ simple on } n \text{ vertices},\; \kappa^{\max}_{G_0} \le m-1 \,\big\}.$$

Proof. By Lemma A.43 feasibility reads $\mu(u,v) \le m - 1 - \pi(u,v)$ on every edge, so the best choice is $\mu(u,v) = m - 1 - \pi(u,v)$, which is what the paragraph before Theorem A.47 records. On an edge $\kappa_{G_0}(u,v) = 1 + \pi(u,v)$, since the routes of G_0 between adjacent u and v are the single edge together with the routes of length at least two. Substituting turns $m - 1 - \pi(u,v)$ into $m - \kappa_{G_0}(u,v)$ and the total into $W_m(G_0)$.

Two things have to be checked for that maximum to be attained by a multigraph over G_0 itself. The first is that the underlying graph of a feasible multigraph is feasible. On an edge, $\kappa_{G_0}(u,v) = 1 + \pi(u,v) \le \mu(u,v) + \pi(u,v) = \kappa_G(u,v) \le m-1$, since $\mu(u,v) \ge 1$. On a non-adjacent pair, $\mu(u,v) = 0$ and $\kappa_{G_0}(u,v) = \pi(u,v) = \kappa_G(u,v) \le m-1$. The second is that the chosen multiplicities are all at least one, so the underlying graph is G_0: $\kappa_{G_0}(u,v) \le m-1$ on an edge gives $\pi(u,v) \le m-2$, hence $\mu(u,v) = m - 1 - \pi(u,v) \ge 1$. Conversely any feasible multigraph is one of the multigraphs considered over its own underlying graph, so nothing is lost by maximising over G_0 first. □

The objective is a sum over edges of a quantity that depends only on the block of the edge. This gives the next result.

Theorem A.50 (The multigraph incidence problem is a knapsack over blocks)**.** *For a 2-connected graph B, or a single edge, write*

$$g_m(b) \;:=\; \max\big\{\, W_m(B) \;:\; B \text{ 2-connected or an edge}, |V(B)| = b,\; \kappa^{\max}_B \le m-1 \,\big\},$$

with $g_m(b) := -\infty$ when no such B exists. Then for all $m \geq 2$ and $n \geq 2$,

$$K_m(n) \;=\; \max\Big\{ \sum_i g_m(b_i) \;:\; b_i \geq 2,\; \sum_i (b_i - 1) \leq n - 1 \Big\},$$

the maximum over all finite multisets of block sizes. In particular the value is determined by the finitely many numbers $g_m(2), \ldots, g_m(n)$.

Proof. The objective is additive over blocks. Let G_0 be feasible and let u, v lie in a common block B. Every u–v path of G_0 stays inside B: a path leaving B must leave through a cut vertex, and to return it would have to pass through that same cut vertex a second time, which no path does. So $\kappa_{G_0}(u, v) = \kappa_B(u, v)$, and since every edge lies in exactly one block,

$$W_m(G_0) \;=\; \sum_{\text{blocks } B} W_m(B).$$

Each block inherits feasibility, $\kappa_B^{\max} \leq \kappa_{G_0}^{\max} \leq m - 1$ by the same identity, so $W_m(B) \leq g_m(|V(B)|)$.

The sizes fit. Suppose G_0 has c components and blocks $B_1, \ldots, B_t$. Within a single component, contracting the block tree makes $\sum(|V(B_i)| - 1)$ equal to the number of vertices of that component minus one. Isolated components contribute zero to both sides, so summing gives $\sum_i (|V(B_i)| - 1) = n - c \leq n - 1$. With Lemma A.49 this proves $\leq$.

Every multiset is realised. Given blocks $B_1, \ldots, B_t$ with $\sum(b_i - 1) \leq n - 1$, take one common vertex and attach the B_i to it, pairwise disjoint otherwise, leaving any spare vertices isolated. The result uses $1 + \sum(b_i - 1) \leq n$ vertices. Its blocks are exactly the B_i, so by the first paragraph its κ agrees with κ_{B_i} on pairs inside a block and equals at most 1 on pairs split between two blocks, which the shared vertex separates. So the graph is feasible and scores $\sum_i W_m(B_i)$, and choosing each B_i optimal gives $\sum_i g_m(b_i)$. This proves $\geq$. □

So $K_m(n)$ can be computed from 2-connected graphs: compute W_m for each, and solve the knapsack problem.

Theorem A.51 (Thickened complete bipartite blocks)**.** *Let $1 \leq s \leq t \leq m - 1$. Thicken every edge of $K_{s,t}$ to multiplicity $m - s$. The result is feasible, $\kappa^{\max} \leq m - 1$, on $s + t$ vertices with total multiplicity $st(m - s)$. Hanging k such blocks off one shared vertex gives, for $n = k(s + t - 1) + 1$,*

$$K_m(n) \;\geq\; \frac{st\,(m - s)}{s + t - 1}\,(n - 1).$$

Optimised over s and t this rate is $\big(3 - 2\sqrt{2} + o_m(1)\big)m^2$, attained at $t = m - 1$ and $s \sim (\sqrt{2} - 1)m$, against the $\big(\frac{1}{8} + o_m(1)\big)m^2$ of Theorem A.47. The improvement is by a factor $8(3 - 2\sqrt{2}) = 24 - 16\sqrt{2} \approx 1.373$.

Proof. Write S and T for the two sides, $|S| = s$ and $|T| = t$.

The connectivities of $K_{s,t}$. For $u \neq u'$ in S, the t two-step routes through distinct vertices of T are internally disjoint. They give $\kappa(u,u') \geq t$. Every route must also meet T, which is a cut of size t, so $\kappa(u,u') = \pi(u,u') = t$. Longer routes may exist but cannot increase this maximum. Symmetrically $\kappa(v,v') = s$ for $v \neq v'$ in T. For adjacent $u \in S$ and $v \in T$, delete the edge uv. The sets $T \setminus \{v\}$ and $S \setminus \{u\}$ both separate u from v, so $\pi(u,v) \leq \min(s-1, t-1) = s-1$. Choose distinct $u_i \in S \setminus \{u\}$ and distinct $v_i \in T \setminus \{v\}$ for $1 \leq i \leq s-1$, possible because $s \leq t$. The routes u, v_i, u_i, v attain the bound. Thus $\pi(u,v) = s-1$ and $\kappa(u,v) = s$.

Feasibility and count. By Lemma A.43 the multigraph has $\kappa = (m-s) + (s-1) = m-1$ on an edge, and $\kappa = t \leq m-1$ or $s \leq m-1$ on a non-adjacent pair, so nothing exceeds the cap. The count is st edges at multiplicity $m-s$. Equivalently, by Lemma A.49, $W_m(K_{s,t}) = st(m-s)$.

The bouquet. If $s \geq 2$, then $K_{s,t}$ is 2-connected, so Theorem A.50 applies and k copies at one shared vertex are feasible with total $k\,st(m-s)$ on $n = k(s+t-1)+1$ vertices. If $s = 1$, then $K_{1,t}$ is a tree (a single edge when $t = 1$), and after thickening every edge to multiplicity $m-1$, sharing vertices among copies still produces a thickened tree and is feasible directly. Its count is again $k\,t(m-1) = k\,st(m-s)$. Thus the displayed rate holds throughout the theorem's full range $1 \leq s \leq t$.

The optimum. For fixed $s > 1$, the factor $t/(s+t-1)$ is strictly increasing in t, and for $s = 1$ it is constant. Thus some optimum has $t = m-1$, its largest permitted value. Put $s = \alpha m$. The rate becomes

$$\frac{s\,(m-1)(m-s)}{s+m-2} \;=\; \frac{\alpha(1-\alpha)}{1+\alpha}\,m^2 \;+\; O(m),$$

and $h(\alpha) = \alpha(1-\alpha)/(1+\alpha)$ has $h'(\alpha) = (1-2\alpha-\alpha^2)/(1+\alpha)^2$, vanishing at $\alpha = \sqrt{2}-1$, where $h = 3-2\sqrt{2}$. Since $3-2\sqrt{2} > \frac{1}{8}$, the family beats the clique bouquet, by the stated factor. □

In the best choice, one side has about $0.41m$ vertices and the other has $m-1$. So $3-2\sqrt{2} - o(1) \leq K_m(n)/(m^2 n) < 2$ for $m \to \infty$ and $n/m \to \infty$. Whether this ratio has a limit is open.

A.10.2 The directed problem

The splitting of κ into parallel arcs and longer routes also works for directed graphs (Lemma A.52). Corollary A.53 below gives the exact value at $m = 2$ and the leading term for every fixed m.

Lemma A.52 (Splitting the directed vertex measure)**.** *Let D be a multidigraph with multiplicity function μ and underlying simple digraph D_0. For an ordered pair $u \neq v$, write $\pi(u,v)$ for the largest number of pairwise internally disjoint directed u–v paths of length at least two in D_0. Then $\kappa_D(u,v) = \mu(u,v) + \pi(u,v)$, and consequently*

$$K_m^{\mathrm{dir}}(n) \;\leq\; (m-1)\,k_m^{\mathrm{dir}}(n).$$

Proof. A directed u–v route of length one is a parallel copy of the arc uv and has empty interior. For the lower bound, take the $\mu(u,v)$ copies of the arc and a maximum family of $\pi(u,v)$ internally disjoint directed u–v paths of length at least two in D_0, lifted into D by choosing one copy of each

arc. Two lifted paths cannot share an arc without sharing an interior vertex, and a path of length at least two cannot use the arc uv, so all these routes are pairwise separated. For the upper bound, a family of separated routes contains at most $\mu(u, v)$ routes of length one, since these are distinct copies of the arc. Its longer routes map injectively to directed paths of D_0, because two routes with the same vertex sequence share an interior that is non-empty, and their images are pairwise internally disjoint, so there are at most $\pi(u, v)$ of them. Note that a directed u–v path uses no copy of the arc vu, so the reverse multiplicity plays no part.

For the bound, let D be feasible. The identity gives $\mu(u, v) \le \kappa_D(u, v) \le m - 1$ on every ordered pair, so the total multiplicity is at most $(m - 1)|A(D_0)|$. And D_0 is itself feasible: on an arc, $\kappa_{D_0}(u, v) = 1 + \pi(u, v) \le \mu(u, v) + \pi(u, v) \le m - 1$, and on a pair with no arc from u to v, $\kappa_{D_0}(u, v) = \pi(u, v) \le m - 1$ directly. So $|A(D_0)| \le k_m^{\mathrm{dir}}(n)$ by definition. □

Corollary A.53 (The directed multigraph incidence value)**.** *For every $n \ge 2$,*

$$K_2^{\mathrm{dir}}(n) \;=\; k_2^{\mathrm{dir}}(n) \;=\; \ell_2^{\mathrm{dir}}(n) \;=\; \max\Big(2(n-1),\; \Big\lfloor \frac{n^2}{4} \Big\rfloor\Big),$$

and for each fixed $m \ge 2$,

$$(m-1)M(n) \;\le\; K_m^{\mathrm{dir}}(n) \;\le\; (m-1)\Big\lfloor \frac{n^2}{4} \Big\rfloor + 4(m-1)^2(n-1),$$

so that $K_m^{\mathrm{dir}}(n) = (m-1)n^2/4 + O_m(n)$ and the bipartite digraph of Theorem 1.13 is asymptotically optimal for this variant too.

Proof. At $m = 2$, two parallel arcs would themselves give two internally disjoint routes, so every feasible multidigraph is simple. The exact value therefore follows from Theorems 1.7 and 1.8. For general m, Theorem 1.13 and $\kappa \le \lambda$ give the lower bound, while Theorem 1.14 at $r = 2$ gives the upper bound. Since $M(n) = \lfloor n^2/4 \rfloor$ for $n \ge 7$, these bounds imply the stated asymptotic. □

At $n = 3$ and $m = 6$, the thickened complete digraph of Construction A.54 has 24 arcs, more than $(m - 1)M(3) = 20$. It is feasible, because every route of length at least two passes through the third vertex. So for the vertex problem the arc construction is not optimal.

Construction A.54 (Thickened complete digraph, and its bouquet)**.** For $2 \le b \le m$, take the complete digraph on b vertices and give every arc multiplicity $m + 1 - b$. At $b = 2$ this is one bidirected edge at multiplicity $m - 1$. Hanging k such blocks off one shared vertex, pairwise disjoint otherwise, uses $k(b - 1) + 1$ vertices, and blocks of different sizes may be mixed.

Proposition A.55 (The directed incidence value is a block problem too)**.** *Let D be a multidigraph with underlying simple digraph D_0, and take the blocks of the underlying undirected graph of D_0. Then κ_D agrees with the value measured inside a block on every pair in a common block, and is at most 1 on every pair split between two blocks. Each block of Construction A.54 is feasible and*

carries $b(b-1)(m+1-b)$ *arcs, so*

$$K_m^{\mathrm{dir}}(n) \;\geq\; \max\Big\{ \sum_i b_i(b_i-1)(m+1-b_i) \;:\; 2 \leq b_i \leq m,\; \sum_i (b_i-1) \leq n-1 \Big\}.$$

Proof. For the block structure, let u, v lie in a common block B of the underlying undirected graph. A directed u–v path of D_0 leaves B only by crossing a cut vertex, and returning crosses that same vertex a second time, which no path does. So every u–v route stays inside B. For two vertices that lie in no common block, either no path joins them or a cut vertex on the intervening block chain separates them. Every directed route must pass through that vertex, giving $\kappa_D \leq 1$ there. The two blocks need not themselves share a vertex.

For one block, the internally disjoint directed u–v routes of length at least two in the complete digraph on b vertices have pairwise disjoint interiors drawn from the other $b-2$ vertices, so there are at most $b-2$ of them, and the two-step routes attain that. Hence $\pi(u,v) = b-2$, and Lemma A.52 gives $\kappa_D(u,v) = (m+1-b)+(b-2) = m-1$ on each of the $b(b-1)$ ordered pairs, which sits at the cap. The count is $b(b-1)(m+1-b)$. Attaching the blocks at one shared vertex uses $1+\sum_i(b_i-1)$ vertices, and any spare vertices are left isolated. □

For small n the block construction is larger, and for large n the bipartite construction is larger. At $m = 3$ the best block is a bidirected edge, so the bouquet is the thickened bidirected tree, and the bipartite construction is larger from $n = 8$. At $m = 6$ the best blocks have three or four vertices. The bouquet has 84 arcs at $n = 8$, against 80, and the bipartite construction is larger from $n = 9$.

A.11 The directed hypergraph

The last four variants use directed hyperedges. The first result gives an upper and a lower bound that differ by a factor of four. Theorem 1.14 then closes this gap.

Proposition A.56 (Directed hypergraphs: first bounds)**.** *Let* $n \geq r \geq 2$ *and* $m \geq 2$*. A directed* r*-uniform hyperedge has one tail and* $r-1$ *heads, and a Berge route enters it at the tail and leaves at a head. If every ordered pair of vertices has at most* $m-1$ *hyperedge-disjoint directed Berge routes, then*

$$|E(\mathcal{H})| \;\leq\; \frac{(m-1)\,n(n-1)}{r-1},$$

and, for every $1 \leq \alpha \leq n-r+1$*, there is a feasible directed multihypergraph with*

$$\alpha \left\lfloor \frac{(m-1)(n-\alpha)}{r-1} \right\rfloor$$

hyperedges. This construction is simple when $m-1 \leq \binom{n-\alpha-1}{r-2}$*. For fixed* r *and* m *and large* n *we can take* $\alpha = \lfloor n/2 \rfloor$*, so the maximum is* $\Theta_{r,m}(n^2)$*.*

Proof. Upper bound. A hyperedge with tail u and heads $h_1, \ldots, h_{r-1}$ joins the $r-1$ ordered pairs (u, h_i). If m hyperedges joined the same pair (u, v), they would give m disjoint routes of length

one. So each of the $n(n-1)$ ordered pairs is joined by at most $m-1$ hyperedges. Counting the pairs (hyperedge, joined pair) in two ways gives

$$(r-1)|E| \le (m-1)\, n(n-1).$$

Lower bound. Split V into a set A of α tails and a set B of $\beta := n - \alpha \ge r-1$ heads. Let $k := r-1$, $d := m-1$ and $N := \left\lfloor \frac{d\beta}{k} \right\rfloor$. Number the vertices of B as $0, 1, \ldots, \beta-1$, and read the repeating sequence

$$0, 1, \ldots, \beta-1, 0, 1, \ldots$$

in consecutive blocks of k numbers. The first N blocks are the hyperedges of a k-uniform multihypergraph $\mathcal{G}^*$ on B. Since $k \le \beta$, each block has k different vertices. The blocks use Nk consecutive positions of the sequence, so every vertex appears $\lfloor Nk/\beta \rfloor$ or $\lceil Nk/\beta \rceil$ times. Since $Nk \le d\beta$, every vertex has degree at most d.

Now give every tail $a \in A$ the hyperedges (a, H) for $H \in \mathcal{G}^*$. No vertex of B is a tail, so every route has length one. So for $a \in A$ and $w \in B$ we get $\lambda(a, w) = \deg_{\mathcal{G}^*}(w) \le m-1$, and there are no other routes. The number of hyperedges is $\alpha N = \alpha \left\lfloor \frac{(m-1)(n-\alpha)}{r-1} \right\rfloor$. If $d \le \binom{\beta-1}{k-1}$, we can use Lemma A.38 instead, which gives different hyperedges, so the directed hypergraph is simple. □

Construction A.57 (Bounded outgoing hyperedge count)**.** Let $n \ge r \ge 2$ and $m \ge 2$, and let $d = \min\{m-1, \binom{n-1}{r-1}\}$. Give every vertex d different outgoing forward hyperedges, with any d different sets of $r-1$ heads. This gives a simple directed hypergraph with nd hyperedges, feasible for both separations. If repeated hyperedges are allowed, give every vertex $m-1$ copies of one outgoing hyperedge, for $n(m-1)$ hyperedges in total.

Proof. Every directed route from u starts with a hyperedge whose tail is u. So removing the outgoing hyperedges of u separates u from every other vertex. There are at most $m-1$ of them, so $\lambda(u, v) \le m-1$ for every ordered pair, and so also $\kappa(u, v) \le m-1$. Hyperedges with different tails are different, so the hypergraph is simple when the head sets at each vertex are different. □

For small n this construction can beat the two-layer construction. For example, at $n = m = 6$ and $r = 3$ it gives 30 hyperedges, even in the simple model.

The forward model is one way to orient a hyperedge. In general, a directed r-uniform hyperedge splits its r vertices into a non-empty *tail set* T and a non-empty *head set* H. A route enters at a tail and leaves at a head. The *forward* model has $|T| = 1$, the *backward* model has $|H| = 1$, and the *general* model allows any split. For $r \ge 4$ there are splits with both parts of size at least two. We show that the backward model is the forward model reversed, and that the general model has the same leading term as the forward model.

Lemma A.58 (Backward is the reverse of forward)**.** *Reversing every hyperedge, $(T, H) \mapsto (H, T)$, turns forward hypergraphs into backward ones and keeps the number of hyperedges.*

The reversed hypergraph $\mathcal{H}^R$ satisfies $\lambda_{\mathcal{H}}(u,v) = \lambda_{\mathcal{H}^R}(v,u)$ and $\kappa_{\mathcal{H}}(u,v) = \kappa_{\mathcal{H}^R}(v,u)$. So the backward model has the same extremal values as the forward model.

Proof. Reversal turns every directed u–v Berge path into a v–u path with the same hyperedges and the same inner vertices. So it keeps both kinds of disjointness. □

Proposition A.59 (The general model obeys the forward bound)**.** *Let a directed r-uniform hyperedge be any split into a non-empty tail set T and a non-empty head set H. If every ordered pair has at most $m-1$ hyperedge-disjoint, or at most $m-1$ internally vertex-disjoint, directed Berge routes, then*

$$|E(\mathcal{H})| \;\le\; \frac{(m-1)\,n(n-1)}{r-1}.$$

The construction of Proposition A.56 also works here, so the maximum is $\Theta_{r,m}(n^2)$.

Proof. Fix an ordered pair (u,v). Each hyperedge (T,H) with $u \in T$ and $v \in H$ gives a route $u \to v$ of length one. Different such hyperedges give separated routes, so there are at most $m-1$ of them. A hyperedge (T,H) joins exactly $|T|\,|H|$ ordered pairs. Summing over all pairs gives $\sum_{(T,H)} |T|\,|H| \le (m-1)\,n(n-1)$. Since $|T|+|H| = r$ and both are at least one, $|T|\,|H| \ge r-1$. So $(r-1)|E| \le (m-1)n(n-1)$. □

So both models lie between two bounds that differ by a factor of 4. This fixes the order n^2, but not the leading constant. The rest of this section finds the constant for both models. Their exact values may still differ.

The proof uses the one-step shadow digraph. Step 1 bounds the hyperedges by the shadow arcs. Step 2 shows that the shadow satisfies a two-step budget. Step 3 applies Lemma A.25.

Definition A.60 (One-step shadow)**.** Let $\mathcal{H}$ be a forward directed r-uniform hypergraph on V. Its *shadow* is the simple digraph $R(\mathcal{H})$ on V with an arc (u,v) when some hyperedge of $\mathcal{H}$ has tail u and head v. Let $\operatorname{cod}(u,v)$ be the number of such hyperedges.

Proof of Theorem 1.14. Let $R = R(\mathcal{H})$. It has no loops, since the tail of a hyperedge is not one of its heads.

Step 1: hyperedges and shadow arcs. The $\operatorname{cod}(u,v)$ hyperedges with tail u and head v give that many routes of length one, which are separated. So $\operatorname{cod}(u,v) \le m-1$. Every hyperedge gives exactly $r-1$ arcs of R. So

$$(r-1)\,|E(\mathcal{H})| \;=\; \sum_{(u,v)\in A(R)} \operatorname{cod}(u,v) \;\le\; (m-1)\,|A(R)|.$$

Proposition A.56 used the bound $|A(R)| \le n(n-1)$ here, which loses the factor four. We now bound $|A(R)|$ better.

Step 2: the shadow is budgeted. Fix an ordered pair (u,v). Let T be the set of midpoints counted by $p_2(u,v)$, together with v if $(u,v) \in A(R)$. So $|T| = \mathbf{1}_{(u,v)\in A(R)} + p_2(u,v)$. Every $t \in T$ is a head of some hyperedge with tail u. Make a bipartite graph that joins each $t \in T$ to the

hyperedges with tail u that contain it. Take a *maximal* matching M in it: one to which no edge can be added.

Then $|M| \geq |T|/(r-1)$. Indeed, if $t \in T$ is not matched, every hyperedge containing t is matched, since otherwise we could add an edge. So every $t \in T$ lies in a hyperedge used by M. Each hyperedge contains at most $r-1$ elements of T, so $|T| \leq |M|\,(r-1)$.

Now turn M into routes. If v is matched to e, take the route $u \to v$ through e. If a midpoint x is matched to e, take the route $u \to x \to v$: enter through e, and leave through a hyperedge with tail x and head v (one exists, since $(x,v) \in A(R)$). The entering hyperedges are different, since M is a matching. The leaving hyperedges have different tails, so they are different. An entering hyperedge has tail u and a leaving one has tail $x \neq u$, so they are different too. So the routes share no hyperedge. Their inner vertices are different midpoints, so they are also separated. So

$$\frac{\mathbf{1}_{(u,v)\in A(R)} + p_2(u,v)}{r-1} \;\leq\; |M| \;\leq\; \kappa_{\mathcal{H}}(u,v) \;\leq\; m-1,$$

and R is $(r-1)(m-1)$-budgeted.

Step 3: conclusion. Lemma A.25 with $C = (r-1)(m-1)$ gives $|A(R)| \leq \lfloor n^2/4 \rfloor + 4(r-1)(m-1)(n-1)$. With Step 1,

$$|E(\mathcal{H})| \;\leq\; \frac{m-1}{r-1}\left[\left\lfloor \frac{n^2}{4} \right\rfloor + 4(r-1)(m-1)(n-1)\right] \;=\; \frac{m-1}{r-1}\left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)^2(n-1).$$

For the lower bound, Proposition A.56 with $\alpha = \lfloor n/2 \rfloor$ gives $(m-1)n^2/(4(r-1)) - O_{m,r}(n)$ hyperedges. This construction is feasible for both separations. For fixed $r \geq 3$ and large n it is simple. So the upper and lower bounds have the same n^2 term. □

Two remarks. For small n, the bound of Proposition A.56 is smaller, roughly until $n \approx 16(m-1)(r-1)/3$. The theorem is only better asymptotically. At $r = 2$ the shadow is the digraph itself, and the theorem gives the quadratic term of Theorem 1.13 again.

The general model needs a different Step 2. The proof above used twice that a hyperedge has one tail: leaving hyperedges with different tails are different, and they differ from the entering hyperedges. With several tails this fails. Two midpoints can leave through the same hyperedge, and for $r \geq 4$ one hyperedge can be the entrance for one target and the exit for another. Instead we take a maximal family of routes and ask why no route can be added. Each unserved target is blocked by a hyperedge that is already used, and a used hyperedge has only r vertices.

Theorem A.61 (The general orientation model shares the constant)**.** *Let $r \geq 2$ and $m \geq 2$, and let a directed r-uniform hyperedge be any split (T,H) into a non-empty tail set and a non-empty head set, as in Proposition A.59. Every such multihypergraph $\mathcal{H}$ on $n \geq 2$ vertices with $\kappa^{\max}_{\mathcal{H}} \leq m-1$ satisfies*

$$|E(\mathcal{H})| \;\leq\; \frac{m-1}{r-1}\left\lfloor \frac{n^2}{4} \right\rfloor \;+\; \frac{4(2r+1)(m-1)^2}{r-1}\,(n-1).$$

Since $\kappa \le \lambda$, the same bound holds when $\lambda_{\mathcal{H}}^{\max} \le m-1$. So for fixed r and m the general model has the same leading term $\big(1+o(1)\big)(m-1)n^2/\big(4(r-1)\big)$ as the forward model, under both separations.

With Lemma A.58, all three models have the same leading term.

Proof. Let R be the shadow in the general model: $(u,v) \in A(R)$ when some hyperedge has u among its tails and v among its heads. Let $\operatorname{cod}(u,v)$ be the number of such hyperedges. Tails and heads are disjoint, so R has no loops.

Step 1. As before, $\operatorname{cod}(u,v) \le m-1$. A hyperedge (T,H) gives $|T|\,|H| \ge r-1$ arcs of R. So

$$(r-1)\,|E(\mathcal{H})| \;\le \sum_{(T,H)\in E(\mathcal{H})} |T|\,|H| \;=\; \sum_{(u,v)\in A(R)} \operatorname{cod}(u,v) \;\le\; (m-1)\,|A(R)|.$$

Step 2: the shadow is budgeted. Fix an ordered pair (u,v). Let $\mathcal{T}$ be the set of midpoints counted by $p_2(u,v)$, together with v if $(u,v) \in A(R)$. Call a hyperedge e an *entrance* for $t \in \mathcal{T}$ if u is a tail of e and t a head. Call e an *exit* for a midpoint t if t is a tail of e and v a head. Every $t \in \mathcal{T}$ has an entrance, and every midpoint has an exit. No hyperedge is both an entrance and an exit for the same t, since t cannot be both a tail and a head.

For each target we use one route. For v it is a route of length one through an entrance. For a midpoint t it is the route $u \to t \to v$ through an entrance and an exit. Build a family of such routes, for different targets and with all hyperedges different, adding routes one at a time until no route can be added. Let $\mathcal{F}$ be the result, with M routes, serving the set S of targets, and let U be the set of hyperedges it uses. Then $|U| \le 2M$. The routes share no hyperedge, and their inner vertices are different, so $M \le \kappa_{\mathcal{H}}(u,v) \le m-1$.

Now take $t \in \mathcal{T} \setminus S$. If t had an entrance outside U and, for a midpoint, also an exit outside U, we could add a route for t. This contradicts maximality. So t is a tail or a head of some hyperedge in U. So

$$\mathcal{T} \setminus S \;\subseteq\; \bigcup_{e\in U} (T_e \cup H_e), \qquad \text{and} \qquad |\mathcal{T}| \;\le\; M + \sum_{e\in U} \big(|T_e| + |H_e|\big) \;=\; M + r\,|U|.$$

With $|U| \le 2M$ this gives $|\mathcal{T}| \le (2r+1)M \le (2r+1)(m-1)$. So R is $(2r+1)(m-1)$-budgeted.

Step 3. Lemma A.25 with $C = (2r+1)(m-1)$ gives $|A(R)| \le \lfloor n^2/4 \rfloor + 4(2r+1)(m-1)(n-1)$, and Step 1 gives the bound. The construction of Proposition A.56 also works in the general model, so the n^2 terms agree. □

The constant $2r+1$ only affects the linear term. Together with Lemma A.58, the theorem shows that the forward, backward and general models have the same leading term.

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